JASA ARTICLE

Multiple patterns of competing tone merging in Guangzhou Cantonese: A production and perception study of young speakers

Lishan Li,[1] Wei Lai,[2] and Xiaoying Xu[1,a)]
[1]School of Chinese Language and Literature, Beijing Normal University, Beijing, China
[2]Department of Linguistics, California State University, Long Beach, Long Beach 90840, USA

ABSTRACT:
Previous studies have documented multiple coexisting tone mergers-in-progress in Cantonese, but it remains unclear how these merging categories interact within the tone system and how production and perception jointly evolve in response to this interaction. This study focuses on ongoing mergers involving three low tones (T3, T4, and T6) in Guangzhou Cantonese, asking whether concurrent merging pathways develop independently or form a structured system of interaction. Production and perception data were collected from 55 younger speakers. Acoustic analyses, combined with *k*-means clustering and Generalized Additive Mixed Modeling (GAMM), revealed four distinct production patterns: a T3–T6 merging pattern, a T4–T6 merging pattern, a three-way merging pattern, and a maintained three-way contrast. Correlation analyses showed that advancement in the T3–T6 merging pattern is negatively associated with advancement in the T4–T6 merging pattern, indicating competition between merging pathways. In addition, progress on the T4–T6 merging pattern is positively associated with the emergence of a three-way merging pattern, suggesting that the T4–T6 merging pattern may act as the primary precursor to the three-way merging pattern. In perception, listeners maintained high discrimination accuracy across all tone pairs, although the T3–T6 contrast was consistently the most difficult. This pattern holds regardless of listeners' tone production patterns. Overall, the results highlight the importance of understanding coexisting merging patterns as interacting pathways within a dynamic system and support a production-led pathway of sound change. The trajectory partly diverges from patterns reported for Hong Kong Cantonese, shedding light on how production and perception interact in multi-path tone merging within a speech community.
© *2026 Acoustical Society of America*. https://doi.org/10.1121/10.0047479

(Received 5 December 2025; revised 18 August 2026; accepted 20 October 2026; published online 9 November 2026)
[Editor: James F. Lynch] Pages: 4013–4033

I. INTRODUCTION

Cantonese has six lexical tones (Zhan, 2002), which are conventionally labeled Tone 1 (T1, /55/), Tone 2 (T2, /25/), Tone 3 (T3, /33/), Tone 4 (T4, /21/), Tone 5 (T5, /23/), and Tone 6 (T6, /22/), following Chao's numerical notation system. In this system, "5" represents the highest pitch level and "1" the lowest (Chao, 1930). Among these six tones, three are classified as low tones: Tone 3 (/33/), Tone 4 (/21/), and Tone 6 (/22/). In some analyses, T4 is alternatively transcribed as /11/, reflecting the view that its phonological representation is primarily low in pitch rather than specified for a falling contour (e.g., Zhan, 2002). Given their shared low-pitch register, they are often easily confused with one another in both speech production and perception (Fung and Lee, 2019; Li, 2008; Mok *et al.*, 2013; Ou, 2012). Recent studies suggest that some of these tones are undergoing mergers. Some studies report the merging of T6 and T3 (Fung and Lee, 2019; Li, 2008), whereas others document an ongoing merger between T6 and T4 (Fung and Lee, 2019). These divergent findings may not be contradictory. Rather, they may reflect that multiple merging trajectories are unfolding simultaneously, particularly among younger speakers who typically lead sound change (Kpogo, 2023; Labov, 1990; Zebe-Sheng *et al.*, 2025). This raises a key question: whether these coexisting merging trajectories interact competitively within the tone system, and if so, how such competition shapes the direction and outcome of tone restructuring.

Competitive mergers for segmental contrasts have been documented. Arnold (2015) reports three vowel mergers before /l/ in Youngstown, Ohio. However, how such mergers interact with production and perception and progress through a speech community in real time remains unclear. Competitive tone mergers-in-progress in tone languages are even less explored, despite growing interest in tone change. In this paper, we conduct both production and perception experiments to examine the directions and advancements of these ongoing low tone mergers among young Cantonese speakers in Guangzhou. Specifically, we investigate how coexisting merging (T3–T6 and T4–T6) interact within the tone system, and how such interaction is reflected in the production–perception relationship and individual variation. By focusing on Guangzhou Cantonese (GZC), this study

a)Email: xuxiaoying2000@bnu.edu.cn

provides a test case for how multiple merging trajectories interact within a single tone system, offering new evidence for the dynamics of competing sound changes and their implications for theories of phonological restructuring.

A. Tone mergers in Cantonese

Empirical studies of Cantonese tone change reveal multiple coexisting patterns of ongoing mergers, but their interaction within the tone system remains poorly understood. Many previous studies have reported tone mergers-in-progress in Cantonese, including varieties spoken in Guangzhou, Hong Kong, and Macau. Documented merging pairs include T2–T5, T3–T6, and T4–T6 (Fung and Lee, 2019; Li, 2008; Mok *et al.*, 2013; Wang *et al.*, 2024; Bauer *et al.*, 2003). Particularly relevant to the present study are the three low tones (T3, T4, and T6), which participate in multiple overlapping merging patterns (e.g., T3–T6 and T4–T6), making them a key locus for investigating potential interactions between coexisting mergers-in-progress. The T2–T5 merging pattern is therefore beyond the scope of the present study, which specifically examines how overlapping merging patterns involving a shared tone may interact.

Tone change in Hong Kong Cantonese (HKC) has been extensively investigated. Bauer *et al.* (2003) documented a bidirectional merging process between the High Rising (T2) and Mid-Low Rising (T5) tones in HKC, noting significant phonetic variability between the two. Regarding low tones, Mok *et al.* (2013) found that among the ongoing tone mergers in Cantonese, T4 appears to be more stable than T6, while T3 and T6 exhibit comparable variability. Liang (2017) identified a reduction in the phonetic distance between these low-register tones, while noting that systematic merging in production was still restricted to a minority of participants. Fung and Lee (2019) observed reduced acoustic distinctiveness among the low-register tones: the T3–T6 and T4–T6 pairs exhibit substantial overlap in their acoustic realizations. These findings suggest that tone mergers-in-progress in HKC are characterized by gradient phonetic convergence rather than complete merger. Importantly, both T3–T6 and T4–T6 mergers involve T6, suggesting that this tone occupies a structurally unstable position and may act as a shared target across multiple merging trajectories. This raises the possibility that these coexisting mergers-in-progress are not independent but potentially interacting within the tone system. This instability is further evidenced in disyllabic contexts; Ren and Mok (2025) found that T3 and T6 are particularly vulnerable to contextual variation, where coarticulatory pressures from preceding tones can significantly reduce the phonetic distance between these merging categories. These findings underscore the instability of low-tone contrasts across different phonetic environments, further complicating the patterns of potential ongoing mergers.

Critically, research on GZC remains notably limited compared to HKC. Although GZC is historically related to HKC and shares a highly similar tone system, it remains unclear whether comparable patterns of gradient merger are observed. Early investigations into GZC tone change (Li, 2008; Ou, 2012) suggest that low-register tones, particularly T3, T4, and T6, may be undergoing merger. Li (2008) reported overlapping F_0 realizations and weakened perceptual distinctions between T3 and T6 among younger speakers. Ou (2012) conducted a series of tone discrimination, identification, and production tasks to investigate tone distinctiveness, finding significant perceptual confusion among low-tone contrasts, particularly between T3–T6 and T4–T6. However, in Ou (2012), tone production was assessed qualitatively, relying mainly on perceptual judgments with only minimal acoustic analysis, which raised concerns about the robustness of these findings. Furthermore, given that multiple merging patterns (e.g., T3–T6 and T4–T6) appear to coexist, an unresolved issue is whether these ongoing mergers develop independently or interact within the tone system.

Given their shared historical roots and highly similar phonological systems of GZC and HKC, one might expect synchronic variations in their sound systems to reflect synchronized diachronic changes (Fung and Wong, 2023). However, since the 1980s, the sociolinguistic landscape of Guangzhou has diverged significantly from that of Hong Kong. While Hong Kong has remained a predominantly Cantonese-speaking society, Guangzhou has evolved into a unique bilingual environment shaped by the promotion of Standard Mandarin and large-scale population mobility. These sociolinguistic differences may influence the trajectory of tone change by altering patterns of language use, input, and intergenerational transmission, potentially leading to divergence in the outcomes of this tone merger-in-progress across the two varieties. Against this background, GZC provides a crucial testing ground for examining how multiple merging trajectories unfold and whether their interaction leads to competition or convergence within the tonal system.

Building on these observations, the present study explores unresolved questions regarding tone mergers-in-progress in GZC within this specific sociolinguistic context. Specifically, it remains unclear how coexisting merging pathways, such as T3–T6 and T4–T6, interact: do they compete or facilitate one another? How do these patterns manifest in the acoustic space, and how do they surface in the relationship between individual perception and production? By examining these questions in GZC, a variety that operates under different sociolinguistic conditions than HKC, this study provides a more comprehensive account of the current tone landscape and its developmental patterns.

B. The relationship between production and perception in sound change

The interplay between speech production and perception is fundamental to understanding sound change. From a communication standpoint, speakers must produce signals that are intelligible to listeners, suggesting a natural coupling between the articulatory and perceptual systems. Many models indeed posit such a tight link, viewing production and perception as functionally symmetrical (Grosvald and Corina, 2012). Empirical evidence further supports this view: for example, Zellou (2017) showed that individuals with greater coarticulatory nasalization in production also

demonstrated heightened perceptual sensitivity to nasal cues. Such findings imply that articulatory behavior may shape perceptual tuning, reinforcing the notion of system integration.

Nevertheless, in real-time sound change, production and perception operate in continuous but imperfect synchrony, giving rise to long-standing debate about which system drives change (Ohala, 1993; Thomas and Hay, 2005). A commonly held view is that changes in perception often precede those in production, suggesting that listeners begin to lose phonemic distinctions before these distinctions are fully reflected in their own speech output. Labov *et al.* (1972) termed this the "perception-first" or "near-merger" pattern. Supporting evidence comes from various languages. Bowie (2000) found that speakers in Waldorf, Maryland, had perceptually merged the GOOSE and FOOT vowels, while maintaining some acoustic differences in production. In the context of a tone merger-in-progress, Liang (2017) reported that in HKC, over half of the speakers they investigated showed tone merging in production. However, only a small fraction preserved full perceptual distinctions across the six lexical tones, suggesting a temporal asymmetry between production and perception during tonal change. Law *et al.* (2013) provided a more nuanced explanation for this phenomenon. They identified speakers who could produce T4 and T6 accurately but failed to distinguish them in conscious perception. Crucially, the presence of a Mismatch Negativity (MMN) in these individuals suggests that their brains still maintain a distinct phonological representation for each tone at a pre-attentive level. These findings align with the Ohala (1993) model of sound change, in which listeners' misperception of coarticulatory cues triggers phonological reinterpretation. Within this framework, perceptual ambiguity arising from articulatory variation may catalyze reanalysis, which stabilizes later in production. Kleber *et al.* (2012) further propose that listeners must first recognize a novel variant before they begin to produce it, framing perceptual shift as a prerequisite for articulatory adoption. Together, these findings highlight the critical role of perception in initiating phonological change and provide a foundation for understanding the temporal dynamics between perception and production in sound change.

However, the reverse pattern, in which production exhibits merging while perception maintains distinctions, has also been documented, though less frequently. For instance, Janson (1983), in his study of the /a:/–/o:/ vowel shift in Swedish, found that younger speakers exhibited greater acoustic shifts in their productions than in their perceptual boundaries. This pattern suggests that these speakers may begin to produce acoustically shifted forms while still largely maintaining older perceptual distinctions. Similar trends have been observed in sociolinguistic studies on different vowel mergers in American English. Baranowski (2013) reported that for the PIN-PEN merger in Charleston, South Carolina, production changes were more advanced than perceptual ones. Likewise, Thomas and Hay (2005) investigated the ELLEN-ALLAN merger among younger New Zealand speakers. They found that while most participants displayed a merger in production, all were still largely able to exploit the distinction for the purposes of speech perception. This was explained by the speakers' continued exposure to distinct variants in their linguistic environment, suggesting that perceptual sensitivity can persist even after production shifts. A parallel pattern is found in the Hailu Hakka tone system: Yeh and Lin (2015) reported that young speakers often produced level tones as falling tones, while still retaining accurate perceptual distinctions between the two tone contours. This indicates that articulatory constraints or motoric economy, rather than perceptual confusability, can drive production-led change. This production-led trajectory is further supported by behavioral timing patterns in Cantonese. Mok *et al.* (2013) demonstrated that while speakers might maintain high perceptual accuracy despite merging in production, this stability often masks an underlying increase in processing effort, as evidenced by significantly protracted reaction times (RTs). This phenomenon was also corroborated by Ou and Law (2016) using event-related potentials (ERPs) to investigate the rising tone contrast (T2–T5). They identified a group of speakers who exhibited compromised production alongside intact perception; crucially, these individuals not only displayed increased discrimination latencies that mirror the behavioral patterns in Mok *et al.* (2013), but also showed distinct neural signatures compared to those with stable perception. Collectively, these findings suggest that the initial stages of a production-led merger manifest as a heightened processing effort, where the maintenance of phonological categories requires greater temporal and neurological resources before any overt decline in perceptual performance occurs.

It is also important to recognize that production-led and perception-led changes can coexist in the same speech community at the same time. Freeman (2023) illustrates this complexity in Pacific Northwest English, where the LEG-VAGUE merger followed a perception-first trajectory, whereas BAG-raising proceeded through production-led mechanisms. A parallel complexity is observed in the tone system of HKC. Fung and Lee (2019) found that while the T4–T6 merger-in-progress exhibited a higher degree of perceptual confusion (aligning with a perception-led model), the T3–T6 merging process exhibited a production-led trajectory, where speakers showed significant acoustic overlap despite maintaining perceptual distinctions. It should be noted that Fung and Lee (2019) characterized these categorizations as primarily descriptive to denote the direction of asymmetry; even for tone pairs identified as merging, participants maintained performance levels significantly above chance. These findings demonstrate that the relationship between perception and production in sound change is dynamic and context-dependent, and that the direction and sequencing of change may vary across linguistic environments.

Overall, this body of research demonstrates that the link between perception and production in sound change is not uniform. While both systems are interdependent, their

evolution can follow different paths across linguistic contexts. The present study explores this relationship in the domain of tone merger in Cantonese, with particular attention to the sequencing and interaction of perceptual and articulatory changes in competing patterns of an ongoing merger.

C. Social and individual factors in language change

Foundational sociolinguistic work systematically demonstrated how macro-level social stratification contributes to sound change. For example, decades of sociolinguistic research have shown that young women frequently lead sound change across a wide range of speech communities (e.g., Labov, 1990, 1994). Women have also been shown to adopt and advance linguistic innovations earlier and at higher rates than men, a pattern observed across both stable and rapidly changing varieties. These trends have been linked to factors such as women's central roles in local social networks, their greater use of socially salient forms, and their participation in emergent identity practices within the community.

Complementing these macro-level patterns, analyzing variation at the individual level is crucial for understanding how these community patterns emerge from the perceptual and articulatory behavior of speakers (Yu and Zellou, 2019; Zhang and Zhu, 2017). To continue with the example of the Thomas and Hay (2005) investigation into the ELLEN-ALLAN merger in NZE, their findings revealed variation across individuals and social groups: although most participants exhibited vowel merging in production, they were still largely able to distinguish them in perception. The authors suggest that individuals who produce merging variants but retain perceptual distinctions may rely on alternative perceptual cues, illustrating how production and perception can diverge within speakers. The importance of individual-level analysis is particularly evident in Cantonese sound change research. For instance, Cheng *et al.* (2022) examined three long-standing consonant merging processes in HKC. While their aggregate results suggested that the community as a whole maintained distinctions in production, the authors explicitly emphasized that these group-level patterns obscured substantial individual variability. Specifically, many speakers had already reached complete merger in both production and perception. This underscores how the link between perceptual and articulatory behaviors is best captured through individual variability, especially when community-level data appear deceptively stable.

Consequently, understanding language change requires bridging these two scales. When individual-level divergences between perception and production are amplified across a community, the resulting landscape is not a simple linear shift, but a complex, multi-pathway process. Crucially, such variability can manifest as co-existing trajectories of change, giving rise to the phenomenon of competing tone-merging trajectories, where multiple tonal categories vie for the same acoustic space.

D. Competitive mergers and their complexity

While much of the literature has traditionally focused on binary contrasts involving only two phonemes, phonological merger is often a more intricate process. The individual and social variations discussed previously frequently give rise to scenarios where multiple phonemes converge simultaneously along different trajectories within the same speech community. These situations lead to what are known as competing mergers (Arnold, 2015; Thomas and Bailey, 1992), a phenomenon typically characterized by significant intra- and inter-speaker variability. Such developments may proceed in parallel or even in opposing directions.

Several studies illustrate this complexity. Arnold (2015) documented overlapping vowel mergers before /l/ in Youngstown, Ohio, where younger speakers tended to exhibit merging between the FOOT and GOAT vowels, while the older FOOT-GOOSE merger showed signs of retreat. A minority of younger speakers further exhibited an apparent three-way merging pattern, though their productions usually patterned with the GOOSE vowel. Likewise, in a study of Scottish English, Lawson *et al.* (2013) demonstrated that vowel merging patterns before postvocalic /r/ varied by social class and articulatory configuration. Middle-class speakers produced a bunched /r/ that led to strong coarticulatory effects and centralization of the KIT, DRESS, and STRUT vowels, while working-class speakers showed a different pattern involving a fronted /r/ and merger of the KIT and STRUT vowels into a low back vowel. Johnson (2007) also examined low vowel mergers in Southeastern New England and found distinct regional trajectories: the THOUGHT and LOT vowels were merging in Boston, while the FATHER and LOT vowels merged in Providence, with younger speakers showing an increasing degree of merging across all three vowels. These findings suggest that the directionality of these ongoing mergers is not always uniform and may involve multiple competing pathways shaped by the complex interplay of region, social factors, and articulatory constraints.

Despite the tonal nature of Cantonese and its complex inventory, comparable investigations of competing tone mergers-in-progress remain scarce. Specifically, research has yet to fully explore how individual-level perception-production patterns contribute to these multi-directional shifts. Existing research has largely described pairwise tone changes, focusing on isolated merging pairs such as T6 (/22/) and T4 (/21/), or T6 (/22/) and T3 (/33/) (e.g., Fung and Lee, 2019; Wang *et al.*, 2024). However, by treating these as independent binary processes, we may overlook the dynamics of the low-tone space as a whole, where T3, T4, and T6 may be simultaneously vying for the same acoustic territory. The coexistence of multiple merging trajectories and the systematic relationship between perception and production in such complex environments remain under-examined.

https://doi.org/10.1121/10.0047479

II. RESEARCH QUESTIONS

Given the complex linguistic landscape of Guangzhou, characterized by increased bilingualism and population mobility, the GZC tone system provides an ideal site for examining sound change in progress. Within this system, the low-register tones are of particular interest due to their high density and acoustic proximity.

Previous studies suggest that T6 (/22/) is variably shifting toward either T4 (/21/) or T3 (/33/). This dual trajectory makes T6 a crucial testing ground for examining the dynamics of competing merging pathways. However, it remains unclear whether these trajectories coexist within the same community, how consistently they are progressing across individuals, and whether perception and production evolve in tandem or along independent paths. In particular, existing work has largely focused on tone merging at the group level, leaving individual variability and the ongoing merger patterns it reveals underexplored.

To investigate these issues, we conducted a monosyllabic tone production task and an AX discrimination task with native Cantonese speakers aged 18–32 from Guangzhou. Acoustic analysis of F_0-based features was used to assess the directionality and advancement of tone merger-in-progress in production, while perceptual discrimination scores provided insights into whether and how tone distinctions are maintained in perception. This approach allows us to capture transitional states and competing pathways in an ongoing sound change. We address the following research questions:

RQ1: Does T6 exhibit multiple competing merging trajectories (i.e., T6→T4, T6→T3, or both), and how are these distributed across individual speakers?
RQ2: How do production and perception interact in these trajectories? Specifically, are the two correlated, mirroring each other across stages of change, or do they evolve asynchronously?

III. METHOD

A. Participants

Sixty participants were recruited, with the majority of them coming from South China Normal University, South China Agricultural University, and Beijing Normal University (30 males, 30 females; age range: 18–32, Mean = 23.62, standard deviatoion, $SD = 3.18$). After excluding three male and two female participants due to poor recording quality (e.g., excessive background noise), data from the remaining 55 participants were retained for analysis. All participants were native speakers of the Guangzhou variety of Cantonese and Mandarin Chinese. None of them had received formal musical training, nor did any report a history of speech, language, or hearing disorders. Participants first completed a production task, then a perception task, and received compensation upon completion. Detailed demographic information for all participants is provided in Table V (Appendix A).

B. Production task

1. Word list

Target syllables were selected from the *Cantonese Pronunciation Dictionary* (广州话正音字典) (Zhan, 2002). The selection focused on syllables that can accommodate all three target tones to form a three-way tonal contrast (T3, T4, and T6). A total of 19 syllables were identified, yielding 57 monosyllabic target words. The complete word list can be found in Table VI in Appendix B.

2. Recording

The production experiment was conducted in a sound-attenuated booth. Target items were presented in random order on the screen, interspersed with filler items. Each trial began when participants pressed the "record" button on the screen and ended when they pressed "stop." Participants were allowed to rerecord each item multiple times until satisfied, and their final recording was retained for analysis. Recordings were captured at a 22.05 kHz sampling rate with a 16-bit resolution using the recording software xRecorder (Xiong, 2017). Prior to the experiment, participants were given written instructions in Simplified Chinese and completed practice trials to ensure task familiarity. During the task, participants were allowed to take breaks as needed. The total duration of this task was approximately 20 min.

3. F_0 extraction and normalization

A semi-automatic annotation script was used to segment the syllables, followed by manual boundary adjustment. F_0 contours were extracted for each syllable using a script with automatic pitch error detection, and were sampled at 11 equally spaced time points. The first and last sampling points were removed to minimize potential edge effects, leaving nine time points for normalization and subsequent analyses. Tokens with unrecoverable pitch tracking errors, which may result from severe creaky voice, recording issues, or other sources of tracking failure, were completely excluded from further analysis. In total, 308 tokens (9.82%) were excluded from the initial 3135 recorded items. Among these excluded tokens, 247 were T4, 42 were T6, and 19 were T3. Given the balanced design (1045 tokens per tone), the exclusion rates were 23.6% for T4, 4.0% for T6, and 1.8% for T3, respectively. The higher exclusion rate for T4 is consistent with its phonetic tendency toward creaky phonation and associated F_0 tracking difficulties. This filtering process yielded a final dataset of 2827 retained tokens.

The retained F_0 values were log-transformed and normalized using the log z-score method (Zhu, 2004), defined as

$$z_i' = \frac{y_i - m_y}{s_y}, \tag{1}$$

where y_i is the logarithmic F_0 value, m_y is the mean logarithmic F_0 across all tones for a given speaker, and s_y is the corresponding standard deviation. This procedure has proven

effective in normalizing F_0 variation across speakers and biological sexes (Zhu, 2004).

Outliers were further screened using a ±3 SD criterion within each tone category for each speaker. Individual F_0 points exceeding this threshold were removed. Remaining isolated missing F_0 points after outlier removal were linearly interpolated within each speaker's pitch trajectory. This interpolation affected only a marginal proportion of the analyzed F_0 points: 0.50% (36 out of 7182 points) for T4, 1.12% (101 out of 9027 points) for T6, and 0.86% (79 out of 9234 points) for T3.

C. Perception task

1. Stimuli

The perception task employed the same 57 monosyllabic target words used in the production task. To generate the auditory stimuli, the words were presented to the speaker in written form, one at a time, in a randomized order. Stimuli were recorded by a 22-year-old female broadcast-speech major, whose pronunciation was judged as clear and representative by the first author, with both being native GZC speakers. Recordings were captured using the xRecorder software at a sampling rate of 22.05 kHz and a 16-bit resolution. Following quality control by the first author, all target syllables were duration-normalized to 320 ms and intensity-normalized to 65 dB.

2. Procedure

The perception task followed an AX discrimination paradigm implemented in Praat (version 6.4.05) with an MFC-7 script (Boersma, 2001). Each trial consisted of target syllables. All possible pairwise combinations of the three tones (T3, T4, and T6) were included using four pairing types: AA, BB, AB, and BA. Each pair was presented once, yielding 228 trials (57 syllable pairs × 4 combinations). An inter-stimulus interval (ISI) of 1000 ms was inserted between the two syllables in each trial. Participants judged whether the two syllables were the same or different by selecting the corresponding option on the screen. No feedback was provided.

3. Calculation of discrimination accuracy score

To account for response bias, a discrimination accuracy score (P) was calculated for each participant and each tone pair using the following formula (Xu *et al.*, 2006):

$$P(\text{obt}) = P(\text{“}S\text{”}|S) \cdot P(S) + P(\text{“}D\text{”}|D) \cdot P(D), \quad (2)$$

where $P(\text{“}S\text{”}|S)$ refers to the percentage of Same responses of all the same pairs, whereas $P(\text{“}D\text{”}|D)$ refers to the percentage of Different responses among all the different pairs. P(S) and P(D) represent the probabilities of same (AA, BB) and different (AB, BA) trials, respectively, and were both set to 0.5 in this experiment. P scores range from 0 (perfectly inaccurate) to 1 (perfectly accurate).

Prior to analysis, data cleaning was performed. First, during stimulus inspection, the syllable "jun" was removed from the dataset, as its pronunciation by the speaker lacked sufficient phonetic distinctiveness and could potentially confound participant judgments. Second, perception data were screened based on RTs: extreme trials (e.g., RT > 10 s) were first excluded, followed by removal of data points exceeding 3 standard deviations from each participant's individual mean RT. After cleaning, a total of 11 187 trials were retained for the final analysis.

IV. RESULTS

In this section, we first present an acoustic analysis of the stimuli from the production task, based on group-level results. We then report the perception accuracy. Finally, we calculate the F_0 height differences among the three tones for each participant and conduct a cluster analysis on these values to identify individual patterns of tone production change. Statistical analyses were conducted using Python version 3.9 (Python Software Foundation, 2020), and the R statistical environment version 4.3.0 (R Core Team, 2023). K-means clustering was performed using the `scikit-learn` package in Python (Pedregosa *et al.*, 2011). Linear and logistic mixed-effects models were fitted using the `lme4` package (v. 1.1–35.5; Bates *et al.*, 2015), while generalized additive mixed models (GAMMs) were implemented via the `mgcv` package (Wood, 2017). Multinomial logistic regression was conducted using the `nnet` package (Venables and Ripley, 2002). *Post hoc* comparisons were performed using the `emmeans` package (Lenth, 2025). Data visualization was primarily performed using the `ggplot2` package version 3.5.1 (Wickham, 2016) within R, and supplemented by custom Python scripts.

A. Production patterns

To visualize the pitch realization of each tone production across participants, we plotted the normalized F_0 contours for all valid production trials in Fig. 1. Each thin line in the figure represents an individual token, while the thick line corresponds to the mean F_0 contours of tone categories T3, T4 and T6.

As shown in Fig. 1, T3 contours are primarily located in the upper region of the F_0 space and exhibit relatively level trajectories. T6, characterized as a low-level tone, occupies the mid-to-lower region of the F_0 space and exhibits a stable level trajectory with a slightly falling slope. T4 tokens are located in the lowest region of the F_0 space, maintaining an exceptionally low pitch trajectory. It should be noted that the residual irregularities observed in some T4 and T6 trajectories are associated with the phonetic characteristics of low-register tones (i.e., creaky phonation in T4 and low F_0 values in T6). These irregularities are limited to a small number of trajectories and are unlikely to substantially affect the mean contours. Although the mean F_0 contours of the three tones remain largely distinguishable, individual tokens show substantial overlap in the F_0 space, especially

https://doi.org/10.1121/10.0047479

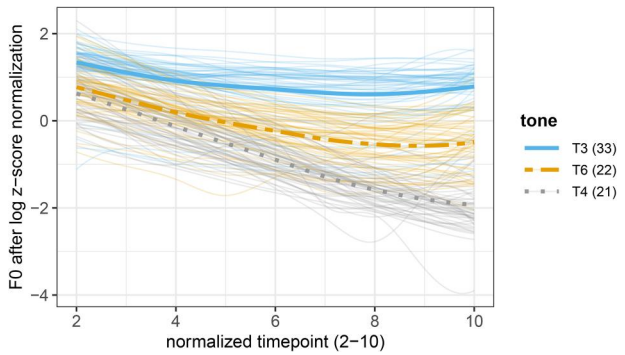


FIG. 1. Log z-score normalized F_0 contours for all tokens of T3, T4, and T6.

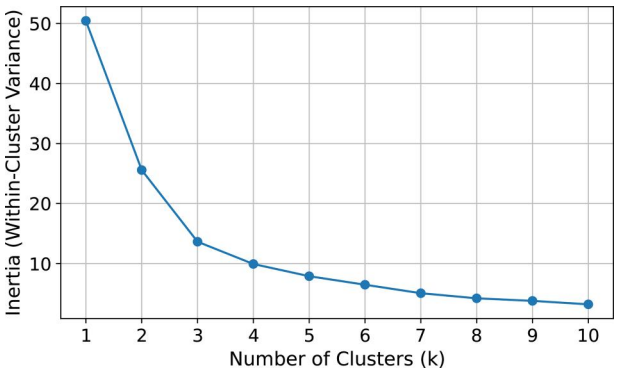


FIG. 2. Elbow plot for choosing optimal K-means clustering.

between T3 and T6, and between T6 and T4, with some overlap also observed between T3 and T4 at the beginning of the tone. The F_0 range of T6 productions is notably wide, spanning from mid-level F_0 values approaching those of T3, to descending contours resembling those of T4. This pattern indicates a high degree of phonetic variability in the realization of the three tones and a potential reduction of phonetic separation between tone categories at the individual level.

To quantify the extent to which different tone categories are merging for each individual, we computed the average F_0 difference (Δ) between each tone pair (T3–T6, T6–T4, and T3–T4) across nine equidistant sampling points (i.e., time point 2–10, excluding onset and offset). Formally, for each tone pair (A, B), this is defined as

$$\Delta_{A-B} = \frac{1}{n}\sum_{i=1}^{n}(F_{0,A,i} - F_{0,B,i}), \tag{3}$$

where $F_{0,A,i}$ and $F_{0,B,i}$ denote the F_0 values of tones A and B at sampling point i, and $n = 9$ is the number of sampled time points. All F_0 values entering this computation are obtained from productions of a single participant, such that Δ_{A-B} summarizes that participant's degree of separation between the two tones. This formulation corresponds to computing the pointwise F_0 difference between the two tones at each sampled time point and then averaging these differences across time.

This yields three indices of ongoing tone merging for each participant, Δ_{T6-T4}, Δ_{T3-T6}, and Δ_{T3-T4}, which respectively capture the extent of tone merging between each tone pair. Since we consistently subtract the lower-pitched tone from the higher-pitched tone, a negative pitch difference would indicate that the two tones have completely merged and no longer maintain the expected high-low distinction, rather than having swapped their acoustic positions. Therefore, we set any values below 0 to denote a full merger. Smaller Δ values indicate a greater degree of merging between tones.

With these metrics, we can cluster participants into different categories, depending on how far advanced they are in the merging process between T6–T4, T3–T4, and T3–T6 in production. To do that, we applied k-means clustering to categorize participants according to their tone production patterns. K-means clustering is an unsupervised machine learning algorithm that partitions data into k clusters by minimizing within-cluster variance. The algorithm iteratively updates cluster centroids until convergence. To determine the optimal number of clusters, we used the Elbow Method, which evaluates inertia (i.e., within-cluster sum of squares) as a function of k. The relationship between k and inertia is illustrated in Fig. 2.

In an elbow plot, the optimal number of clusters is typically found at the point where adding more clusters yields diminishing returns; in other words, where the curve bends and the decrease in inertia begins to level off. By examining the second-order differences in inertia across various k values, we identified the optimal cluster number as four, resulting in four distinct groups based on their ongoing merging patterns in production.

To visualize the clustering results, we constructed a three-dimensional (3D) scatterplot (Fig. 3). The three axes represent pairwise tonal distances derived from acoustic measurements: Δ_{T3-T6} (x-axis), Δ_{T6-T4} (y-axis), and Δ_{T3-T4} (z-axis), corresponding respectively to the distances between tones /33/ and /22/, tones /22/ and /21/, and tones /33/ and /21/ in Cantonese. Each dot represents one participant, identified by their numeric ID label. The participants are grouped into four clusters using K-means clustering, with each cluster visualized in a different color.

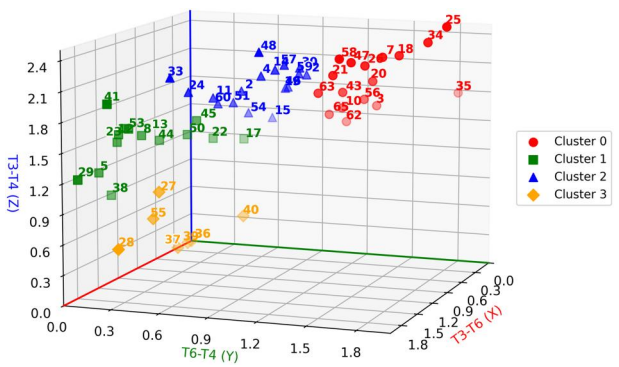


FIG. 3. K-Means Clustering (four clusters) with participant IDs.

In Fig. 3, participants are divided into four clusters. Cluster 3 (yellow) is clustered around the origin (0, 0, 0) and includes those with low values in all three dimensions of Δ_{T3-T6} (X-axis), Δ_{T6-T4} (*Y*-axis), and Δ_{T3-T4} (*Z*-axis). For those participants, /33/, /22/, and /21/ exhibit an advanced degree of merging. In contrast, all other participants show high values for Δ_{T3-T4}, meaning that their /33/ and /21/ remain distinct. Cluster 1 (green), located near the XZ plane, is characterized by low Δ_{T6-T4} but relatively high Δ_{T3-T6}, suggesting that /22/ and /21/ exhibit greatly reduced acoustic separation while /33/ and /22/ remain distinct. Cluster 0 (red), located near the YZ plane, shows the opposite pattern, with low Δ_{T3-T6} but relatively large Δ_{T6-T4}: these participants took the opposite route in the competition between the T3–T6 and T4–T6 merging pathways by favoring the T3–T6 route. Cluster 2 (blue) occupies an intermediate position between Clusters 0 and 1: these participants do not align closely with either the XZ plane (T4–T6 merging pattern) or the YZ plane (T3–T6 merging pattern) but instead maintain distinctive distances among all three tone pairs. Notably, no points fall in the XY plane, indicating that a T3–T4 merging pattern never occurs without prior merging involving either T3–T6 or T4–T6.

Table I presents the cluster assignment for each speaker group, along with the number and proportion of participants in each group, based on the clustering results shown in Fig. 3.

Figure 4 presents the results of K-means clustering as 3 two-dimensional (2D) projections (XY plane, YZ plane, XZ plane) of the 3D feature space depicted in Fig. 3. Again, the color codes reflect the same four clusters identified by the K-means algorithm (Clusters 0–3), and data points represent individual participants with associated numeric labels indicating participant IDs.

Pearson correlations were conducted to examine the relationships among the three indices of tone merging. In the XY-plane projection, the correlation between Δ_{T3-T6} and Δ_{T6-T4} was negative ($r = -0.34$, $p = 0.011$), which is likely driven by the opposite trajectories taken by participants in Cluster 1 versus Cluster 0. This indicates that participants who show a smaller distance (i.e., greater degree of merging) for T3–T6 tend to maintain a larger distance (i.e., lesser degree of merging) for T6–T4, and vice versa. The negative association suggests that advancement along one merger dimension is accompanied by resistance along the other, lending support to the idea that the two merging processes constitute competing pathways in the reorganization of the tone system. In the YZ-plane projection, the correlation between Δ_{T6-T4} and Δ_{T3-T4} was strongly positive ($r = 0.74$, $p < 0.001$), indicating that participants who are merging T6 and T4 are also more likely to show subsequent merging between T3 and T4. Last, a weaker positive correlation was observed between Δ_{T3-T4} and Δ_{T3-T6} ($r = 0.34$, $p = 0.01$), suggesting some association, though less robust. Taken together, the negative correlation in the XY-plane supports the interpretation that the T3–T6 merging pattern and the T6–T4 merging pattern are competing pathways. The other two projections highlight the subsequent merger between Δ_{T3-T4}: since Fig. 3 demonstrated that T3–T4 does not merge independently of the other pairs, a low Δ_{T3-T4} value is an indicator of a three-way merger. The correlations further suggest that while either a T3–T6 or T6–T4 merging pattern can lead to a three-way merging pattern, the T6–T4 merging pattern is a stronger predictor of this outcome.

TABLE I. Cluster assignment, group size, and proportion of participants.

Cluster	Pattern	Number	Proportion
0	T3–T6 merging pattern	17	30.91%
1	T4–T6 merging pattern	14	25.45%
2	Three-way contrast	17	30.91%
3	Three-way merging pattern	7	12.73%

Based on the clustering results, the four groups were categorized to illustrate the diverse synchronic variation in F_0 contours, as plotted in Fig. 5.

Figure 5 displays the group-level normalized F_0 trajectories for the three target tones, T3 (/33/), T6 (/22/), and T4 (/21/), across four clusters of speakers. The figure reveals distinct patterns of tone realizations and varying degrees of ongoing tone merging in each cluster. Cluster 2 serves as the baseline group, exhibiting clearly differentiated tone categories. The three tones maintain distinct F_0 contours across the time course. This indicates that the three-way tonal contrast is largely maintained in this group. Cluster 1 presents an ongoing merger between T4 and T6, as characterized by their overlapping F_0 contours and similar shapes. Notably, the F_0 contour of T6 falls more steeply than in Cluster 2 and increasingly approximates the contour of T4, indicating a merging process of T6 toward T4. In contrast, Cluster 0 exhibits an opposite merging pattern. In this cluster, T3 and T6 both display relatively flat contours, and the pitch-height spacing between them is compressed, especially toward the end compared to Clusters 1 and 2, indicating reduced separation between the two tones. By comparison, T4 retains its characteristic extreme-low pitch realization and remains well separated from the other two tones. Cluster 3 reflects the most advanced pattern of tone merging, where all three tones show substantial overlap in the acoustic space, although T4 still retains a slightly steeper F_0 contour than T3 and T6. This pattern indicates a high degree of overlap among the three tonal categories, consistent with an advanced stage of three-way tone merger-in-progress.

While the K-means clustering approach identifies speakers with similar merging profiles based on their overall acoustic distance among tone pairs, it does not directly characterize the temporal evolution of tonal trajectories. Therefore, we further applied GAMMs to statistically evaluate whether the identified clusters exhibited systematic differences in their F_0 trajectories and to provide a dynamic characterization of the merging patterns revealed by the clustering analysis. To rigorously model the time-course variations and statistical significance of these F_0 trajectories across clusters, GAMMs were fitted to the acoustic data

https://doi.org/10.1121/10.0047479

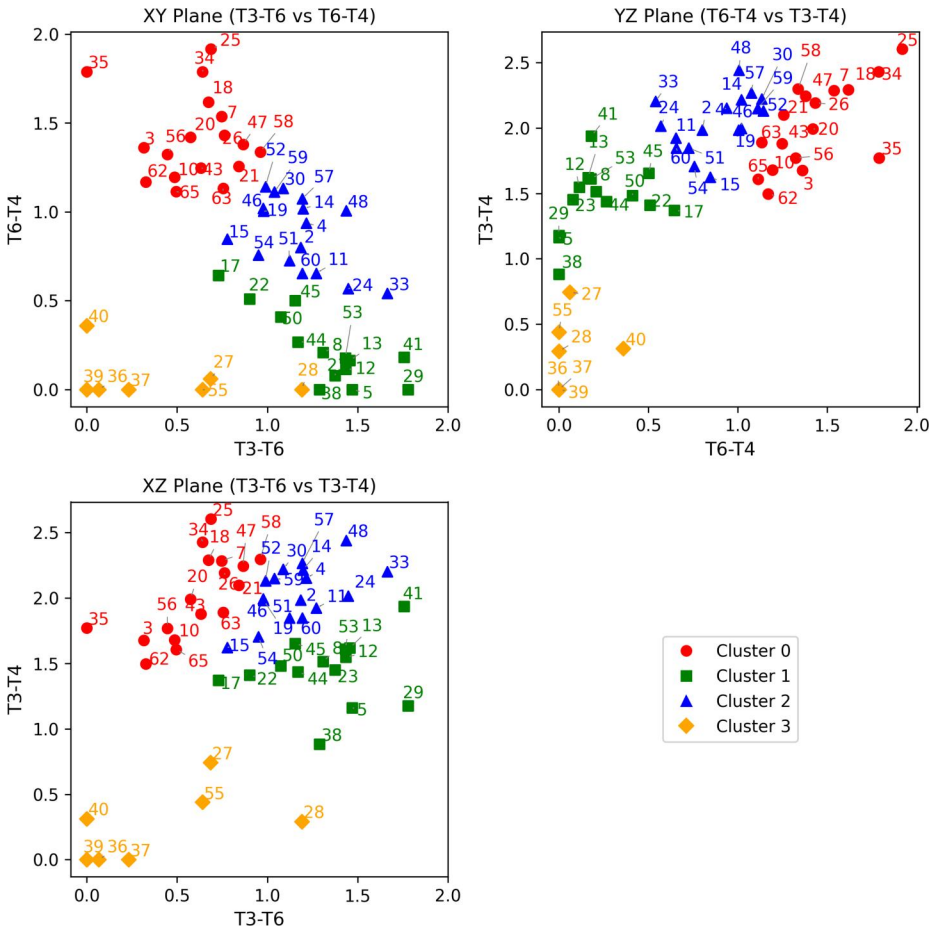


FIG. 4. K-means clustering of participants based on tone distance measures in 3D space, projected onto 2D planes.

using the `bam()` function with the `fREML` method. The dependent variable was z-score normalized F_0. The model incorporated a parametric interaction between `Tone` and `Cluster`, along with cluster-specific tone trajectory smooths (`by = ToneCluster`, $k = 5$) to capture the nonlinear temporal trajectories associated with each tone–cluster combination. To account for speaker-specific differences in pitch range and contour shape, as well as item-specific variability, random factor smooths for `Participant` (`bs="fs"`) and random intercepts for `Character` (`bs="re"`) were included. Furthermore, to prevent type I error inflation caused by temporal autocorrelation inherent in dynamic time-series acoustic tracking, a first-order autoregressive (AR1) error model was applied. A relatively high autocorrelation parameter ($\rho = 0.90$) was adopted to account for strong temporal dependency in F_0 trajectories, following previous dynamic speech-acoustic GAMM applications. Figure 6 (row 1) displays the normalized F_0 trajectories for T3 (/33/), T6 (/22/), and T4 (/21/), which were estimated by zeroing out the random effects to reveal the

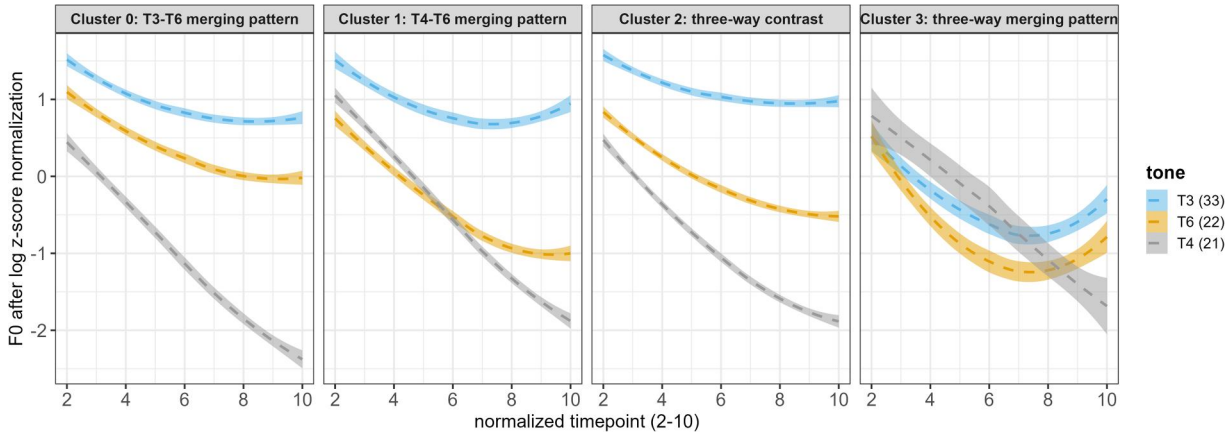


FIG. 5. Production patterns of four clusters of participants.

cluster-level trajectories associated with each cluster while controlling for participant- and item-level variability.

To locate the exact time windows where the merging clusters deviate from the non-merging reference group (Cluster 2), pairwise difference smooths were generated (rows 2–4 of Fig. 6). Instead of relying on *post hoc* marginal estimations, the statistical significance of these deviations and their 95% confidence intervals (CIs) were computed directly via the linear predictor matrix (*lpmatrix*) and the full variance-covariance matrix of the fitted GAMM (Wieling, 2018), ensuring rigorous control over inter-condition covariances. The results reveal distinct, statistically significant deviation patterns across the syllable rhyme (indicated by red shaded regions where the confidence CI strictly excludes zero, $p < 0.05$). For instance, the convergence in Cluster 1 (T4–T6 merging pattern) appears to involve bidirectional acoustic adjustments: T4 is significantly raised in the early-to-mid rhyme [Fig. 6(G)], while T6 is significantly lowered in the mid-to-late rhyme [Fig. 6(H)]. Conversely, Cluster 0 (T3–T6 merging pattern) is primarily associated with T6 showing significantly higher F_0 values throughout the time course (Panel E), whereas T3 remains statistically undifferentiated from the baseline [Fig. 6(F)]. Cluster 3, reflecting the three-way convergence pattern, displays substantial deviations across all three tones, featuring a significantly raised T4 alongside markedly lowered T6 and T3 [Figs. 6(J)–6(L)].

While the single-tone GAMM comparisons illustrate localized phonetic deviations, establishing a trajectory of an ongoing tone merger requires demonstrating a systematic reduction in the acoustic contrast space between tone pairs. To this end, following the methodological framework for dynamic phonetic comparisons outlined by Wieling (2018), a dynamic difference smooth analysis was conducted within

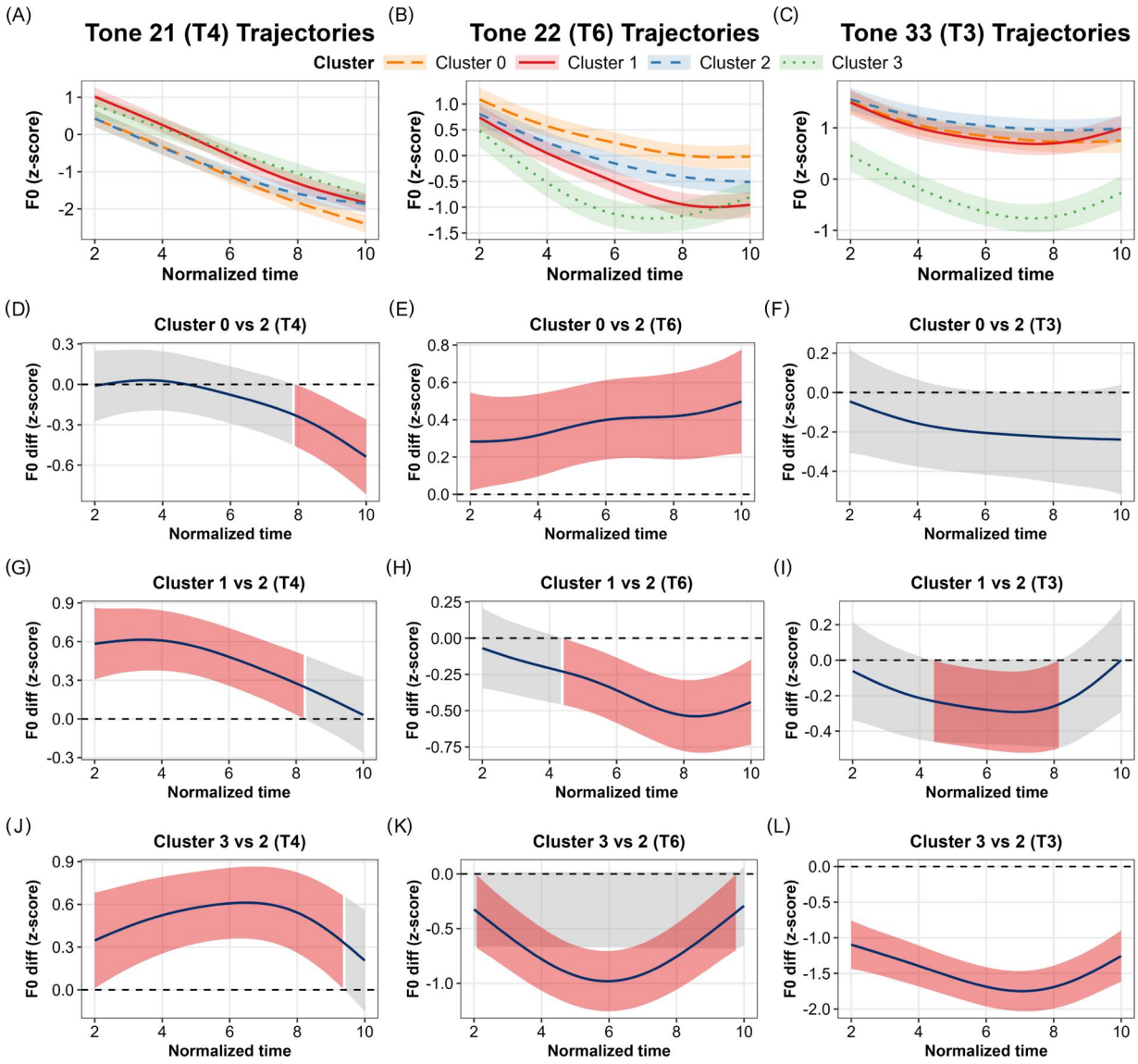


FIG. 6. GZC tonal trajectories and pairwise cluster differences. Row 1 displays population-level F_0 trajectories with 95% CI. Rows 2–4 present pairwise GAMM difference smooths comparing each target cluster to the non-merging baseline (Cluster 2) for T4, T6, and T3. Red shaded regions indicate statistical significance ($p < 0.05$) where the CI does not overlap with zero.

https://doi.org/10.1121/10.0047479

the GAMM framework (Fig. 7). By computing the time-varying acoustic contrast between two tones within a target cluster, with the acoustic contrast calculated as the canonically higher-F_0 tone minus the canonically lower-F_0 tone to maintain a positive baseline contrast, and subsequently subtracting the corresponding tonal contrast observed in the non-merging reference group (Cluster 2), this approach quantifies time-varying changes in tonal contrast. Formally, this dynamic difference smooth (represented as ΔC_{F_0} on the y-axis) is defined as

$$\Delta C_{F_0} = (F_{0,\text{higher}} - F_{0,\text{lower}})_{\text{Target Cluster}} - (F_{0,\text{higher}} - F_{0,\text{lower}})_{\text{Reference Cluster}}. \quad (4)$$

Consequently, a negative value indicates that the tone pair is closer together in the target cluster than in the reference baseline (i.e., acoustic contrast compression), whereas a positive value indicates contrast expansion. To avoid inflated type I error rates associated with pointwise comparisons, standard errors for the difference smooths were estimated using matrix operations on the model *lpmatrix* and the full covariance matrix following Wieling (2018).

As illustrated in Fig. 7, the difference smooth trajectories reveal distinct patterns of acoustic contrast reduction that correspond closely to the merging profiles identified by the K-means clustering. Specifically, the tone-pair relationships captured by the clustering based on global ΔC_{F_0} contrast differences are further characterized by the GAMM-based dynamic analyses, which demonstrate that these contrast reductions are not static differences but unfold systematically across the tonal trajectory. For example, Cluster 0, corresponding to the T3–T6 merging pattern, shows a significant negative deviation in the T3–T6 acoustic contrast [Fig. 7(B)], indicating a reduction in the acoustic contrast between these two tones over the tonal trajectory. Meanwhile, the positive values in the T6–T4 contrast [Fig. 7(A)], suggest that this merging pattern is accompanied by a redistribution of acoustic space rather than a generalized compression of all tonal contrasts. Cluster 1 exhibits a different pattern, with a significant reduction in the T6–T4 contrast across nearly the entire normalized time course [Fig. 7(D)], consistent with the T4–T6 merging profile identified by the clustering analysis. At the same time, changes in the T3-related contrasts occur in a time-dependent manner, with an increased relative T3–T6 separation toward the end of the contour [Fig. 7(E)] and reduced T3–T4 separation across the majority of the contour [Fig. 7(F)]. In contrast, Cluster 3 displays a global reduction in tonal contrasts, with significant negative differences in

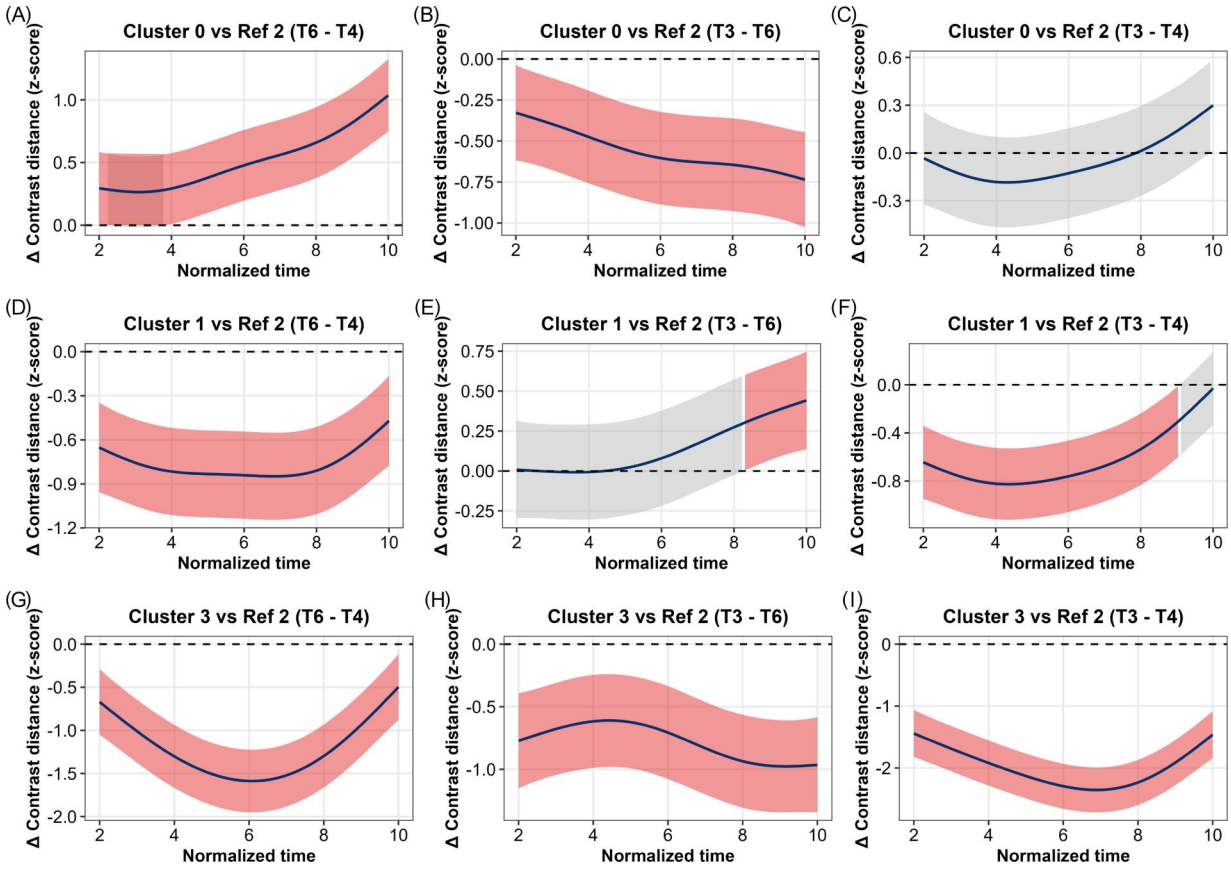


FIG. 7. Dynamic difference smooths evaluating tonal contrast reduction across time. The 3×3 matrix compares the acoustic contrast of specific tone pairs (T6–T4, T3–T6, T3–T4) in Clusters 0, 1, and 3 against the non-merging reference Cluster 2. Red shaded regions indicate statistically significant deviations ($p < 0.05$). Specifically, negative values indicate that the corresponding tone-pair contrast is reduced relative to the reference cluster, whereas positive values indicate contrast expansion.

multiple tone pairs, including T6–T4 [Fig. 7(G)], T3–T6 [Fig. 7(H)], and T3–T4 [Fig. 7(I)]. This widespread contrast compression corresponds to the extensive overlap among the three tonal categories observed in this cluster.

Although the group-level GAMMs and difference smooth analyses establish robust population-wide trends, it is crucial to verify that these statistical patterns are consistently reflected at the individual level, rather than being driven by a subset of extreme outliers. To evaluate the consistency of these patterns, the normalized F_0 contours for each of the 55 participants are presented in Fig. 8, faceted by their assigned cluster. In each facet, thin lines represent individual tokens, while the bold lines indicate predicted mean contours estimated via participant-level GAMMs using cubic shrinkage splines ($k = 5$). The *X*-axis represents normalized time across nine equidistant sampling points. To preserve the relative acoustic contrast between tone categories while accommodating significant inter-speaker variation in baseline pitch range, the *Y*-axis employs independent scales (*free_y*) for each participant.

This granular visualization further demonstrates that the speaker-specific trajectories are broadly consistent with the group-level statistical patterns. For instance, individuals in Cluster 3 (e.g., P27, P40) consistently exhibit overlapping contours for all three tones, providing qualitative evidence of consistent overlap across tokens for a three-way merging pattern. In contrast, participants in Cluster 2 (e.g., P2, P11) maintain clear, parallel separation throughout the duration of the tones, serving as a stable baseline.

To examine whether `Sex` affects the distribution of these ongoing merger patterns among participants, a multinomial logistic regression model was fitted. The dependent

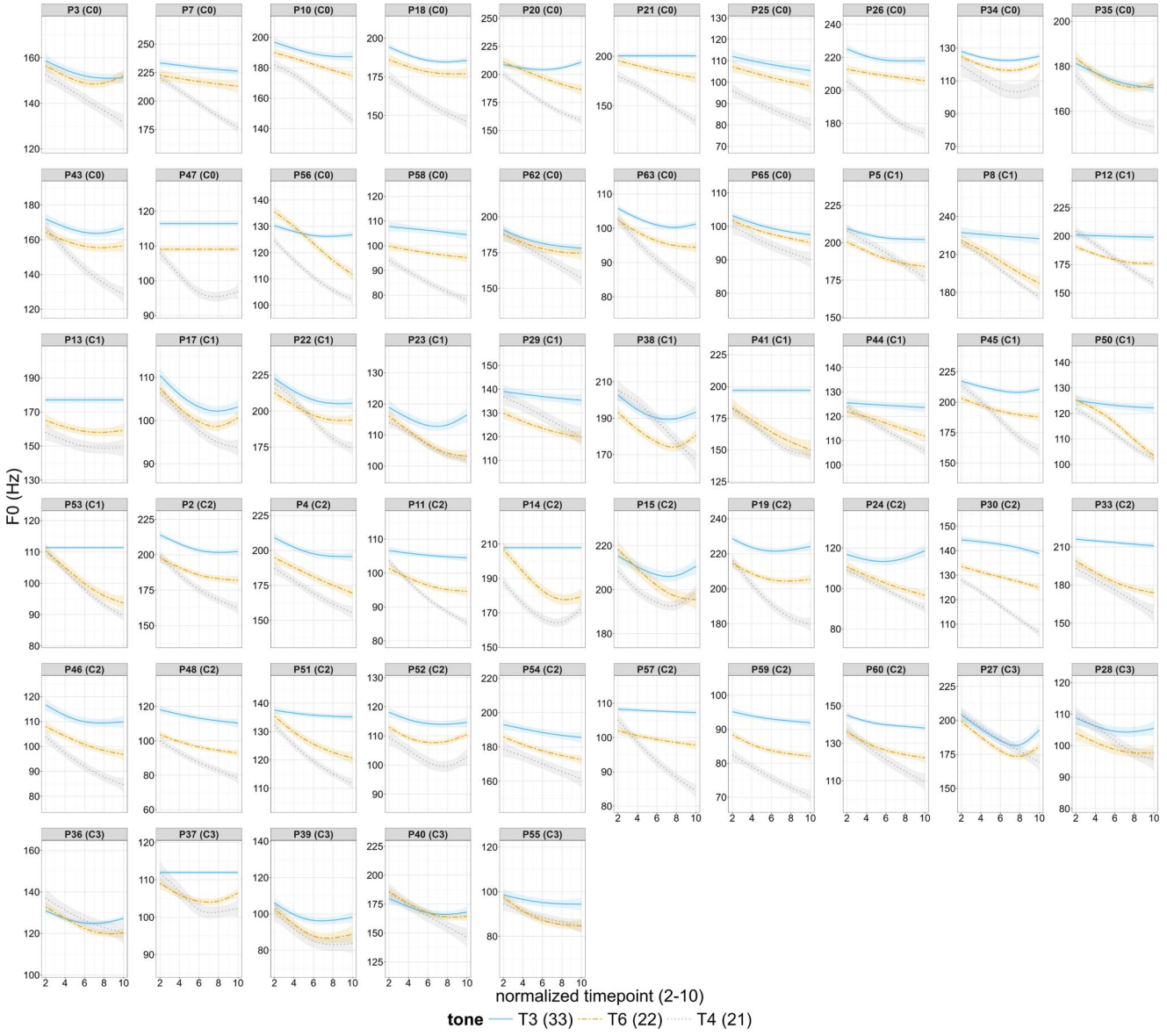


FIG. 8. Individual F_0 contours for all 55 participants, grouped by cluster membership (C0–C3), where numeric labels (prefixed with P) indicate unique participant IDs. Thin lines represent individual tokens, and thick lines indicate the mean contours estimated by GAMMs for T3 (solid blue), T6 (dashed orange), and T4 (dotted gray).

https://doi.org/10.1121/10.0047479

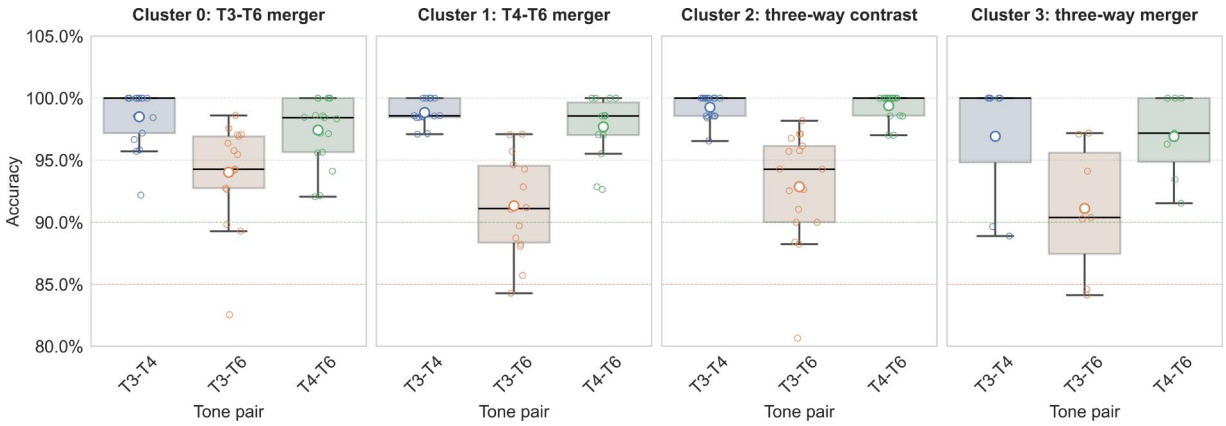


FIG. 9. Perception patterns of four clusters of participants.

variable was `Cluster` (four patterns, coded as 0–3, baseline: `Cluster0`), with `Sex` (`female` vs `male`) as the predictor.

The model results indicated no significant effect of `Sex` on cluster membership. The estimates and 95% CI for the `Sex` effect (`female` vs `male`) were as follows: `Cluster1` vs `Cluster0`: $\beta = -0.069$, 95% CI [−0.896, 0.758]; `Cluster2` vs `Cluster0`: $\beta = -0.713$, 95% CI [−1.502, 0.075]; `Cluster3` vs `Cluster0`: $\beta = -0.644$, 95% CI [−1.673, 0.384]. All 95% CIs included zero (all $p > 0.05$), providing no evidence of a significant association between `Sex` and cluster membership. Thus, male and female participants showed comparable distributions across the ongoing merging patterns.

B. Perception results

To visualize the perceptual accuracy for each cluster, we plotted the distribution of perception accuracy (%) for the three tone pairs (T3–T4, T3–T6, T4–T6) across the four participant clusters (Fig. 9). Each panel represents a distinct participant cluster, with its hypothesized phonological pattern noted in the panel title (e.g., "Cluster 0: T3–T6 merging pattern"). The *Y*-axis represents accuracy (in percent), and the *X*-axis represents the three tone pairs. Within each boxplot, the box spans the interquartile range (IQR; 25th–75th percentiles); the thick horizontal line indicates the median; the large open circle represents the mean accuracy; whiskers extend to the most extreme data points within 1.5 times the IQR; the small, jittered dots represent the average accuracy of individual participants; and horizontal dashed lines indicate 85%, 90%, 95%, and 100% accuracy levels for visual reference.

As shown in Fig. 9, the mean perceptual accuracy for all participants across all tone pairs was above 80%. Overall, T3–T4 was the most easily distinguished pair, whereas T3–T6 was the most perceptually challenging, a difficulty that was particularly pronounced in Cluster 1 and Cluster 3. Cluster 0 (T3–T6 merging pattern) showed extremely high accuracy on both T3–T4 and T4–T6 (medians both > 97%), but exhibited a noticeable drop in accuracy for the T3–T6 condition (approximately 94%), along with greater individual variability. Cluster 1 (T4–T6 merging pattern) displayed a similar but more extreme pattern: participants maintained high accuracy on T3–T4 and T4–T6 (medians both > 97%), but performed worst on T3–T6 (median approximately 90%), showing greater dispersion. This pattern suggests that the perceptual distinction between T4 and T6 remains relatively robust despite substantial production overlap. Cluster 2 (three-way contrast) demonstrated consistently high perceptual accuracy across all three tone pairs (all medians > 93%), especially on T3–T4 and T4–T6, where accuracy approached 100%. In contrast, Cluster 3 (three-way merging pattern) performed best on T3–T4 (median near 100%), followed by T4–T6 (approximately 95%) and T3–T6 (approximately 90.5%).

To statistically assess the influence of cluster membership, tone pair, and sex on perceptual accuracy, we fitted a logistic mixed-effects regression model to predict listeners' `Response` (Same = 0, Diff = 1) on each Trial, with `Cluster` (dummy-coded, baseline: Cluster0), `Sex` (dummy-coded, baseline: Female), `Tone Pair` (dummy-coded, baseline: T3–T4), and their three-way interaction as the fixed effects, and `Participant` and `Syllable` as random intercepts. Model selection was carried out through backward stepwise comparisons using `anova()` to determine the optimal model fit. Starting from a maximal model, we sequentially removed one predictor at a time, beginning with higher-order interaction terms and moving to lower-order ones. Within the same level, insignificant predictors were evaluated before significant ones. If the model comparison resulted in a value of $p < 0.05$, the model that provided the better fit was chosen. Otherwise, if the two models were not significantly different, the simpler model was chosen.

The model selection procedure first removed the three-way interaction term `Cluster:Sex:TonePair` [$\chi^2(2) = 10.30$, $p = 0.11$], and then the two-way interaction term `Cluster:Sex` [$\chi^2(3) = 3.42$, $p = 0.33$]. The remaining items were preserved because they significantly improved the model fit. The final model included the main factors of `Cluster` (dummy-coded, baseline: Cluster0), `Sex` (dummy-coded, baseline: Female), `TonePair` (dummy-coded, baseline: T3–T4), the two-way interactions

TABLE II. Results of the mixed-effects model: *Response* ~ *Cluster* * *TonePair* + *Sex* * *TonePair* + (1|*Participant*) + (1|*Syllable*). *Note*. SE = standard error. Significance codes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Predictor	Estimate	SE	*z*	p
(Intercept)	3.29	0.28	11.71	<0.001***
Cluster1	−0.51	0.26	−2.00	0.046*
Cluster2	−0.20	0.25	−0.79	0.430
Cluster3	−0.40	0.34	−1.18	0.238
TonePair T3–T4	1.95	0.36	5.44	<0.001***
TonePair T4–T6	1.22	0.28	4.33	<0.001***
Sex (Male)	−0.21	0.20	−1.05	0.292
Cluster1:TonePair T3–T4	0.72	0.43	1.68	0.094.
Cluster2:TonePair T3–T4	1.08	0.47	2.30	0.022*
Cluster3:TonePair T3–T4	0.50	0.54	0.92	0.356
Cluster1:TonePair T4–T6	0.57	0.34	1.68	0.094.
Cluster2:TonePair T4–T6	1.80	0.46	3.92	<0.001***
Cluster3:TonePair T4–T6	0.79	0.47	1.67	0.095.
TonePair T3–T4:Sex (Male)	−0.88	0.36	−2.42	0.016*
TonePair T4–T6:Sex (Male)	−0.62	0.30	−2.10	0.036*

`Cluster:TonePair` and `Sex:TonePair`, as well as random intercepts for `Participant` and `Syllable`, to predict listeners' `Response` (baseline: Same) on each Trial. The result of this model is shown in Table II.

The statistical results, shown in Table II, revealed a significant effect of `Cluster1` ($\beta = -0.51$, $p = 0.046$), indicating that participants with a `T4-T6` merging pattern had a lower accuracy overall compared to participants with a `T3-T6` merging pattern, which is consistent with Fig. 9. As for TonePair, the model revealed significant effects for both `T3-T4` ($\beta = 1.95$, $p < 0.001$) and `T4-T6` ($\beta = 1.22$, $p < 0.001$). These results indicate that compared to the tone discrimination performance on tone pair `T3-T6`, stimuli of both `T3-T4` and `T4-T6` induced higher accuracy in tone discrimination. This is also consistent with Fig. 9, which shows the lowest accuracy rate for `T3-T6`.

Moving to interaction terms, the model showed significant interaction effects between `Cluster2` and `TonePairT3-T4` ($\beta = 1.08$, $p = 0.022$), as well as the interaction between `Cluster2` and `TonePairT4-T6` ($\beta = 1.80$, $p < 0.001$). This means that compared to `Cluster0`, participants of `Cluster2` achieved a higher accuracy both in discriminating stimuli of `T4` vs `T6` and in discriminating stimuli of `T3` vs `T4`. This is expected since participants in `Cluster2` maintained a three-way contrast in production, which might contribute to their higher accuracy in perceptual discrimination. Finally, the model showed significant interactions between `SexMale` and `TonePairT3-T4` ($\beta = -0.88$, $p = 0.016$) as well as `TonePairT4-T6` ($\beta = -0.62$, $p = 0.036$). This means that for these two tone pairs, which generally exhibited higher discrimination accuracies, a sex difference was observed such that female listeners exhibited higher discrimination accuracy than male listeners.

To examine pairwise differences in model effects, *post hoc* pairwise comparisons of estimated marginal means were conducted using the `emmeans` package (Lenth, 2025). We first compared TonePairs within each cluster, and then compared clusters within each tone pair.

Table III presents the pairwise comparisons of tone pairs within each of the four clusters, calculated as `log odds ratios`. The results consistently demonstrate that across all four clusters, the `T3-T6` pair shows a significantly lower log odds ratio compared to both the `T3-T4` pair and the `T4-T6` pair. Specifically, the contrast `T3-T6` and `T3-T4` pairs yielded highly significant negative estimates in all clusters (`Cluster0`: $\beta = -1.51$; `Cluster1`: $\beta = -2.24$; `Cluster2`: $\beta = -2.59$; `Cluster3`: $\beta = -2.01$; all $p < 0.001$). A similar significant negative difference was found for the `T3-T6` and `T4-T6` contrast (`Cluster0`: $\beta = -0.91$; `Cluster1`: $\beta = -1.48$; `Cluster2`: $\beta = -2.71$; `Cluster3`: $\beta = -1.70$; all $p < 0.001$). In contrast, the difference between the `T3-T4` and `T4-T6` pairs failed to reach statistical significance in any cluster (`Cluster0`: $p = 0.142$; `Cluster1`: $p = 0.106$; `Cluster2`: $p = 0.972$; `Cluster3`: $p = 0.845$). This pattern indicates that while

TABLE III. Pairwise comparisons of tone pair within each Cluster (log odds ratio). *Note*. SE = standard error. Significance codes: $^{*}p < 0.05$, $^{**}p < 0.01$, $^{***}p < 0.001$.

Cluster	Contrast	Estimate	SE	df	z.ratio	*p*-value
Cluster = 0:	T3–T6 - T3–T4	−1.51	0.29	Inf	−5.27	<0.001***
	T3–T6 – T4–T6	−0.91	0.23	Inf	−3.90	0.001***
	T3–T4 - T4–T6	0.60	0.32	Inf	1.89	0.142
Cluster = 1:	T3–T6 - T3–T4	−2.24	0.33	Inf	−6.81	<0.001***
	T3–T6 - T4–T6	−1.48	0.25	Inf	−5.95	<0.001***
	T3–T4 - T4–T6	0.76	0.37	Inf	2.02	0.106
Cluster = 2:	T3–T6 - T3–T4	−2.59	0.38	Inf	−6.80	<0.001***
	T3–T6 - T4–T6	−2.71	0.40	Inf	−6.82	<0.001***
	T3–T4 - T4–T6	−0.12	0.52	Inf	−0.23	0.972
Cluster = 3:	T3–T6 - T3–T4	−2.01	0.47	Inf	−4.31	<0.001***
	T3–T6 - T4–T6	−1.70	0.41	Inf	−4.11	<0.001***
	T3–T4 - T4–T6	0.31	0.56	Inf	0.55	0.845

https://doi.org/10.1121/10.0047479

T3-T4 and T4-T6 are statistically indistinguishable from each other in this analysis, both yielded higher discrimination performance than T3-T6, irrespective of cluster membership in the ongoing merging process.

We then examined whether discrimination performance differed across clusters for each tone pair. Pairwise comparisons between clusters revealed that differences were specific to the T4-T6 pair. For T3-T4 and T3-T6, no cluster comparisons reached significance (all $p > 0.05$), suggesting homogeneity across these ongoing merging patterns for these pairs. In contrast, for T4-T6, Cluster2 showed significantly higher log odds ratios than Cluster0 ($\beta = -1.597$, $p = 0.0028$) and Cluster1 ($\beta = -1.540$, $p = 0.0062$), whereas other inter-cluster comparisons were not significant. These results indicate that the clustering solution primarily differentiates performance on the T4-T6 pair, with Cluster2 exhibiting superior discrimination relative to Cluster0 and Cluster1. Detailed results of all *post hoc* comparisons are reported in Appendix C.

To further investigate whether the maintained perceptual accuracy across clusters was achieved at the expense of increased processing effort, we analyzed listeners' RT. A linear mixed-effects model (LMM) was fitted. The model formula was specified as follows:

$$RT \sim Cluster \times TonePair + (1 \mid Participant). \quad (5)$$

In this model, RT was the dependent variable. Fixed effects included Cluster (four levels, baseline: Cluster0), TonePair (three levels, baseline: T3-T4), and their interaction. Participant was included as a random intercept to account for individual variability. The results of the model are summarized in Table IV.

As shown in Table IV, there was no significant main effect of Cluster (all $p > 0.40$), indicating that the overall RTs did not differ significantly across production-based groups. A significant main effect of TonePair was found, with the T3–T6 contrast eliciting significantly longer RTs ($\beta = 0.087, p = 0.002$), suggesting that this pair required more processing effort to discriminate than the T3-T4 pair. Crucially, the absence of a significant Cluster × Tone Pair interaction (all $p > 0.13$) suggests that even speakers with an advanced degree of tone merging (Cluster3) processed these contrasts as efficiently as the stable group (Cluster2).

TABLE IV. Results of the mixed-effects model for RTs: $RTs \sim Cluster \times TonePair + (1|Participant)$. *Note:* T3-T4 and Cluster0 serve as the reference levels. SE = standard error. Significance codes: $^{**}p < 0.01$, $^{***}p < 0.001$.

Predictor	Estimate	SE	*df*	*t*	*p*
(Intercept)	0.694	0.077	55.41	8.969	<0.001***
Cluster 1	0.097	0.115	55.37	0.843	0.403
Cluster 2	−0.068	0.109	55.21	−0.620	0.538
Cluster 3	0.083	0.143	55.37	0.579	0.565
TonePair T3–T6	0.087	0.028	2563.03	3.063	0.002**
TonePair T4–T6	0.011	0.028	2563.01	0.377	0.706
Cluster 1: T3–T6	0.026	0.042	2563.02	0.618	0.537
Cluster 2: T3–T6	−0.027	0.040	2562.88	−0.679	0.497
Cluster 3: T3–T6	−0.079	0.052	2562.86	−1.512	0.131
Cluster 1: T4–T6	0.024	0.042	2563.33	0.558	0.577
Cluster 2: T4–T6	0.035	0.039	2562.85	0.887	0.375
Cluster 3: T4–T6	0.022	0.052	2562.81	0.418	0.676

V. DISCUSSION

This study investigates how Cantonese low tones undergo merging in both perception and production across individuals, and how these two domains interact in the process. Our discussion of the results is organized around three aspects: We first examine the pathways through which ongoing tone mergers unfold in production (Sec. V A), followed by a discussion of participants' perceptual performance that accompanies these shifts (Sec. V B). Finally, we discuss the relationship between production and perception over the course of this ongoing low-tone merging process in GZC, and situate these patterns in comparison with the tone merger-in-progress in HKC (Sec. V C).

A. The pathways of tone merging in production

Our results of acoustic measurements indicate that the ongoing merging of the three low tones T3, T4, T6 among young Cantonese speakers in Guangzhou follows multiple trajectories. Since the three low tones are situated in the mid-to-low register of the Cantonese tone system, mergers among them are phonetically plausible, given their proximity in acoustic space (Qin and Mok, 2013; Qin and Jongman, 2016). Using k-means clustering analysis, we identified four groups of speakers who exhibited distinct patterns of tone production: (1) speakers who are advanced in the T3–T6 merging process, (2) speakers who are advanced in the T4–T6 merging process, (3) speakers who exhibit a three-way merging pattern of T3, T4, and T6, and (4) speakers who maintain relative acoustic distinctions among the three tones. Crucially, as revealed by the single-tone GAMM comparisons and the dynamic difference smooth analysis, these merging patterns are not static shifts in overall pitch height. Rather, they are temporally evolving processes where the reduction of acoustic contrast varies dynamically during the course of production. Our analysis reveals that the different pathways of tone merging are not random but are structurally constrained. Correlational analyses indicate a significant negative correlation between the T3–T6 merging pathways and the T4–T6 mergers, suggesting that speakers who are advanced along one merging trajectory are typically resistant to the other. This aligns with theoretical accounts of sound change as a process of competing variants (Mufwene, 2008). As Blythe and Croft (2012) note, competition among multiple forms is characteristic of sound change in progress. The present study also uncovers synchronic implicational relationships underlying the three-way merging pattern (T3–T4–T6). In our data, a three-way merging pattern only occurs when either a T3–T6

merging pattern or a T4–T6 merging pattern is already in place, since no speaker directly exhibits a merging between T3 and T4 without first showing one of the other two pathways. This is not unexpected, as T3 and T4 are both acoustically more distant and differ in contour shape, making direct merging between them less likely. Instead, a merging process involving T6 appears to act as a precursor that reduces the effective contrast space, thereby facilitating subsequent convergence across all three tones. Furthermore, correlation analyses indicate that the T3–T4 merging process is more strongly associated with T4–T6 ($r = 0.74$) than with T3–T6 ($r = 0.34$), suggesting that the T4–T6 merging pathway is a primary precursor to the three-way merging pattern.

Such patterns, where an initial merging stage sets off a series of subsequent shifts, resemble developments reported in other phonological changes. Our GAMM results demonstrate that such mechanisms operate dynamically in Cantonese: for instance, the T4–T6 merging process involves bidirectional acoustic shifts, while the T3–T6 merging process is driven unilaterally by T6 rising into the T3 space. For example, Gardner and Roeder (2022) document the low back merger shift, in which a three-way merger among LOT-THOUGHT-PALM subsequently triggered chain shifts of TRAP, DRESS, and KIT. Comparable phenomena have also been reported in tone languages (e.g., Min dialect, Zhang and Zhu, 2017). Our results suggest that a similar mechanism may operate in Cantonese: as T6 (/22/) undergoes merging with T4 (/21/), the contrastive function of the height feature between T3 (/33/) and T6 (/22/) may be reduced. Consequently, the distinction between T3 and the merging T6 may become less dependent on height, which could facilitate the development of a three-way merging pattern. This indicates that merging processes need not be limited to linear pairwise processes; rather, they may involve dynamic interactions leading to multi-way restructuring. Speakers who exhibit a three-way merging pattern may represent a more advanced stage of this synchronic restructuring process.

Having established these internal structural relationships (competitive and implicational), we then explored correlations with conventional macro-social factors. This analysis diverges from previous tone-merging studies which often show clear stratification across social groups (Arnold, 2015; Freeman, 2014; Labov, 1990, 1994; Molina García, 2023; Ou, 2012). Interestingly, no consistent sex-related pattern emerged across all tone pairs in our production data, indicating that while sex may influence performance in specific contrasts, the observed variation in sound change patterns appears to be strongly associated with internally structured phonological factors. In perception data, however, sex differences are observed for specific tone pairs (T3–T4 and T4–T6), but the perception patterns do not align with the speakers' advancement in ongoing tone merger, as we will discuss in the next section. This null finding, however, does not necessarily mean that this change lacks social embedding. It is possible that social factors are operating at a more fine-grained level (e.g., social networks, individual identity) that is not adequately captured by the broad male-female distinction used in this study. Taken together, our findings highlight that sound change can unfold through internally structured competition, with implicational relationships among variants guiding merger trajectories even when macro-social stratification is weak or inconsistent.

B. Tone discrimination performance in perception

To investigate how ongoing mergers among the low tones contribute to perceptual ambiguity, we conducted an AX discrimination experiment involving each of the possible tone contrasts, T3–T6, T4–T6, and T3–T4. Despite evidence from the production data showing that low tones are undergoing merger through multiple pathways, the perception results indicate that young Cantonese speakers in Guangzhou exhibit robust perceptual distinctions for those tone pairs: across all participants, mean accuracy for all three tone pairs exceeded 80%, far above chance level of 50%. These findings suggest that while production exhibits signs of an ongoing merger, perception remains stable, pointing to a temporal lag between the two domains, which we discuss in more detail in the next section.

To examine whether perceptual accuracy for different tone pairs varied across participants in the four production-derived clusters, we conducted pairwise comparisons between clusters for each tone pair. Perceptual accuracy for T3–T4 and T3–T6 did not differ significantly across clusters, suggesting broadly similar perceptual abilities despite divergent production patterns. The only notable difference emerged for T4–T6, where participants in Cluster 2 (who maintain a three-way contrast) outperformed those in Clusters 0 and 1. Overall, then, perceptual differences across clusters are minimal and largely restricted to the T4–T6 contrast. We also conducted pairwise comparisons across tone pairs within each participant cluster, with the results showing systematic differences in perceptual distinctiveness across contrasts. *Post hoc* tests consistently showed that T3–T6 was the most difficult to distinguish: across all clusters, discrimination accuracy was significantly lower for T3–T6 compared to either T3–T4 or T6–T4 (all $p < 0.05$). These results confirm that T3–T6 is perceptually the most challenging contrast for young GZC speakers.

The perceptual difficulty between T3 and T6 is not random, but arises from their minimal acoustic distance and the inherently gradient perceptual processes engaged when distinguishing level tones. From an acoustic perspective, T3 (/33/) and T6 (/22/) are both level tones distinguished primarily by subtle F_0 height differences. By contrast, the T4 (/21/) vs T6 (/22/) opposition benefits from more robust acoustic cues: T4 is realized in the lower portion of the speaker's pitch range, creating a more salient F_0 difference than the T3–T6 pair. In addition to this extreme low-pitch target, previous research has shown that T4 in Cantonese is often characterized by creaky voice (Yu and Lam, 2014). This phonation cue may provide additional perceptual support for identifying T4 and distinguishing it from

neighboring tone categories, compensating for the lack of distinct contour differences among these low-register tones. Similarly, the T3–T4 contrast has the largest acoustic distance, incorporating both the maximal F_0 height difference and the distinct phonation cue. This combination of static pitch distance and non-modal phonation makes T3–T4 and T4–T6 much easier to perceive than the purely height-dependent T3–T6 contrast.

Overall, the perception experiment shows that perceptual categorization remains relatively stable across speakers, even where production patterns diverge. This dissociation suggests that perceptual differentiation does not align with speaker-specific merger trajectories and is therefore difficult to reconcile with a strong perception-led account of the observed production patterns. One important factor contributing to this observation is that the perception experiment was based on stimuli produced by a speaker with relatively standard pronunciation and minimal tone merger. As a result, the task primarily reflects listeners' ability to categorize well-formed, canonical tokens, which likely contributes to the consistently high accuracy across participants and does not directly mirror the variability observed in their own productions. However, real-world listening involves a much broader range of input conditions than those represented in the present task. As tone categories begin to converge in production, the acoustic input available to listeners would also become less distinct. This reduction in distinctiveness may introduce perceptual ambiguity downstream, which could in turn reinforce the merging process over time. In this sense, while the present study shows that progressive contrast reduction in production precedes perceptual merging among younger GZC speakers, such progressive contrast reduction in production may eventually contribute to changes in the perceptual system. Future studies can further investigate the possibility of a production-led pathway of change with downstream effects on perception.

C. The relationship between perception and production in the ongoing GZC low tone merging process

Asymmetries between production and perception have long been a central theoretical concern in research on sound change. Some changes are perception-led, with distinctions lost in perception before production (e.g., Bowie, 2000; Labov *et al.*, 1972; Ohala, 1993; Liang, 2017), whereas others are production-led, with changes in production preceding perceptual adjustments (e.g., Janson, 1983; Baranowski, 2013; Yeh and Lin, 2015; Thomas and Hay, 2005). More recent work emphasizes the bidirectionality of sound change, showing that within the same community, different changes may follow different trajectories (e.g., Freeman, 2023).

In the present study, younger speakers of GZC demonstrated robust perceptual discrimination of the T3–T6, T4–T6, and T3–T4 tone pairs, while production data revealed clear evidence of merging tendencies in the majority of participants. Three distinct production merging trajectories were inferred from these clustering patterns. Based on the *k*-means clustering of production data, the 55 younger GZC speakers were stratified into four groups reflecting different degrees of tonal erosion. Specifically, 30.91% of speakers (Cluster 0) showed a T3–T6 merging pattern, 25.45% (Cluster 1) showed a T4–T6 merging pattern, and 12.73% (Cluster 3) produced an advanced three-way merging pattern of all low-register tones, while 30.91% (Cluster 2) maintained a stable three-way contrast. Despite these differences in production, perceptual discrimination remained high, with accuracy exceeding 80% for all tone pairs across all clusters. These results indicate that listeners can distinguish the tone categories even when production contrasts are reduced or absent. The mismatch between production and perception shows that tonal changes in production can occur without immediately affecting perceptual categories, supporting a production-led tendency in the current stage of the GZC low-tone merger-in-progress. In the ongoing T6 merging process, production changes appear to precede perceptual changes, a pattern that is consistent with cases documented in Swedish, American English, and Hailu Hakka (Baranowski, 2013; Janson, 1983; Yeh and Lin, 2015).

This finding deviates partially from previous findings on the production-perception relationship in ongoing tone mergers in HKC, highlighting a clear regional divergence in tone evolution. Fung and Lee (2019) documented complex production-perception asymmetries in HKC. They described the T3–T6 opposition as a case of what may be termed partial merger, characterized by a reduction in production contrast while the perceptual contrast is largely retained. This is consistent with the perceptual stability observed in GZC. However, a crucial regional divergence lies in the T4–T6 opposition. In HKC, T4–T6 exhibits a pattern akin to a near-merger, where perceptual contrast shows signs of erosion earlier while production remains relatively intact. It should be noted that these categorizations are used as analogical descriptions of evolutionary patterns; experimental data indicate that while the distinctiveness of these tone pairs has diminished, it remains significantly above the chance level. Further support comes from Liang (2017), who showed that in HKC, 19% of speakers exhibited "perceptual merger despite production contrast," compared to only 6% who showed the reverse pattern. Together, these studies demonstrate that in HKC, tone change often exhibits patterns consistent with perception-led tendencies, although mixed trajectories are also attested. By contrast, GZC speakers maintain robust perceptual distinctions for both T3–T6 and T4–T6, which is more consistent with a production-led trajectory. Unlike HKC speakers in Mok *et al.* (2013), who showed slower RTs despite perceptual maintenance, younger GZC speakers demonstrated high perceptual accuracy, with no significant RT differences associated with their production-based merging patterns. Results from a linear mixed-effects model (Table IV) revealed no significant main effect of `Cluster` (all *p*-values > 0.40) and, crucially, no significant interaction between `Cluster` and

Tone Pair (all p-values > 0.13). This contrast may indicate that perceptual stability in GZC is not accompanied by the same pattern of RT differences as that reported for HKC. While recent work on both varieties points to directionally mixed patterns (i.e., the coexistence of perception-led and production-led shifts), the consistently production-led pattern and the absence of significant RT differences associated with production-based merging patterns in GZC further support the interpretation of a production-led tendency, compared to the more effortful and variable asymmetries reported for HKC.

While these synchronic data offer strong evidence for a production-led trajectory, longitudinal tracking and experimental manipulations are needed to fully specify the causal link between production weakening and perceptual shifts. One potential explanation for the observed production-perception decoupling is that articulatory divergence may accumulate over time due to contextual variation in natural speech. Although the present study focuses on monosyllabic production, the cluster-specific patterns may reflect the long-term effect of phonetic fluctuations encountered in more complex speech environments. For instance, recent research on disyllabic tones (Ren and Mok, 2025) highlights the role of coarticulation in increasing the susceptibility of low-register tones to merging. Such contextual effects, together with sociolinguistic influences like reduced GZC use in formal domains, may underlie the diverse merging trajectories observed in younger speakers. Future research should investigate how early-degree production changes, potentially via incremental reweighting of auditory categories and exposure to variable speech contexts, eventually precipitate perceptual merger.

Our findings highlight regional differences between tone merger patterns in GZC and HKC. Although GZC has historically served as the main prestige standard for HKC, more than a century of political separation and divergent development may have led to differences between the two. Over the past several decades, the linguistic landscape in Guangzhou has been shaped by two concurrent forces (Chen, 2011; Zhu, 2021). The first is a top-down national policy promoting Standard Mandarin as the official language in high-prestige domains such as education, government, and mass media. The second is urbanization, which brought a large influx of migrants from other provinces, creating a bilingual environment in which Mandarin functions as a lingua franca for communication between local Cantonese speakers and non-Cantonese populations. As a result, Mandarin has become the dominant language in many daily and professional contexts, compressing the functional space of GZC. These factors may particularly affect the younger generation and could be related to the reduced production differentiation observed among younger speakers, especially in formal contexts such as schools. However, these factors are not directly tested in the present study.

The continued vitality of Cantonese in popular culture, including songs, films, and television dramas, together with its ongoing use in family and community interactions among middle-aged and older generations, ensures that younger speakers are still regularly exposed to Cantonese in their daily lives. Such sustained auditory input is likely to play a role in preserving perceptual sensitivity, even in the face of reduced opportunities for production. In contrast, Cantonese in Hong Kong enjoys a highly stable functional status across virtually all domains of daily life. It serves as the primary native language and the most frequently used language in everyday communication. Even newly arrived migrants from mainland China exhibit high proficiency and frequent use of Cantonese, albeit speaking with a strong accent (Sautman and Xie, 2020).

Finally, our results provide new insights that extend previous research on GZC. Among earlier studies on GZC, Li (2008) observed weakening in both production and perception for the T3–T6 contrast, and Ou (2012) reported a dissociation between production and perception. However, much of this work relied primarily on qualitative judgments or limited acoustic analysis, which may have constrained the ability to detect subtle early-stage changes. In contrast, the present study employs fine-grained acoustic measurements, enabling more sensitive identification of production weakening before perceptual distinctions are lost among the younger generation of GZC speakers. Taken together, these findings suggest that production changes precede perceptual shifts in the ongoing tone-merging process among younger GZC speakers, while also highlighting the need for further longitudinal and experimental work to confirm the directionality of change.

D. Conclusion and future work

This study investigated several ongoing tone mergers involving three low tones, T3, T4, and T6 in GZC, by examining individual-level variation in tone production and perception among younger speakers. Our results reveal that this sound change does not follow a single, uniform trajectory. Instead, multiple coexisting production patterns were identified, including speakers who exhibit a T3–T6 merging pattern, a T4–T6 merging pattern, or a complete three-way T3–T4–T6 merging pattern, alongside speakers who still maintain the full three-way contrast. Analyses of the relationships among these pathways suggest a structured, hierarchical pattern: the T3–T6 and T4–T6 merging pathways appear to be in competition, with individual speakers generally exhibiting a stronger tendency toward one pattern than the other. Between the two processes, T4–T6 merging trajectory appears to be more closely associated with the three-way merging pattern, providing insights into how complex, multi-path tone mergers-in-progress unfold within a speech community. From a perceptual perspective, the T3–T6 contrast consistently posed the greatest difficulty in tone discrimination, regardless of speakers' production patterns. While sex was examined as a potential factor in social stratification, its effect on both perception and production appears to be limited. Overall, our data indicate a production-led trajectory for the low-tone merger-in-progress in GZC: progressive contrast reduction in production precedes

perceptual merging among younger GZC speakers. This production-driven pattern is not fully parallel to the tonal evolution reported for HKC, suggesting nuanced differences in the mechanisms of sound change across the two varieties.

More broadly, tone merging is best understood as a multi-stage and multi-dimensional process rather than a single end point. In the present study, we focus on the early stages of this merging process, in which tone categories begin to approach one another in the acoustic space, as reflected in reduced separation in F_0. At the same time, the merging process may also involve changes in other dimensions, such as increasing similarity in contour shape, which may become more or less observable at different stages of the process. The interaction among these dimensions of tonal convergence is itself a non-trivial issue. An important direction for future research is therefore to examine how multiple dimensions of tonal convergence emerge and interact over time, including how reductions in distance and changes in contour shape jointly contribute to the overall merging pattern, and how these changes affect both production and perception at different stages.

Finally, while the present study shows that shifts in production precede perceptual merging among younger GZC speakers, such progressive contrast reduction in production will ultimately create perceptual ambiguity within the system. Our perception data were elicited using speech from a broadcast-speech major student who maintained relatively clear tone distinctions in production; the fact that listeners still experienced perceptual difficulty with these clear tokens further highlights the influence of production-driven shifts on the perceptual system. Future research should therefore examine the perceptual patterns of fully merged speakers to determine whether and when production-based contrast reduction becomes sufficiently robust to reshape perceptual categorization. Tracing this transition will be crucial for understanding how production-led changes propagate through the perceptual system and, ultimately, how tone mergers become stabilized within the speech community.

ACKNOWLEDGMENTS

This research was supported by the National Social Science Fund of China "A Study on the Change Mechanism and model of Chinese Tone Categories, Perceptual and Productive Phonetic Cues" (24BYY121). We are also deeply grateful to all participants for their time and contribution to this study. We also thank the anonymous reviewers for their insightful comments and constructive suggestions, which have significantly improved the quality of this manuscript.

AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts of interest to disclose.

Ethics Approval

This study involved only non-invasive speech production and perception experiments with healthy adult participants. No sensitive or personally identifiable information was collected, and the research posed minimal risk to participants. According to the institutional and national guidelines in China, formal review and approval by the Ethics Committee of Beijing Normal University was not required for this type of non-invasive behavioral research. All participants were fully informed about the purpose, procedures, and potential risks of the study, and provided written informed consent prior to participation. The consent process ensured that participants understood their voluntary participation and their right to withdraw at any time.

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

APPENDIX A: PARTICIPANT INFORMATION

Appendix A provides demographic information about the participants (Table V).

TABLE V. Participant information.

ID	Sex	Age	District	ID	Sex	Age	District
1	F	20	Tianhe	34	M	27	Yuexiu
2	F	23	Liwan	35	F	23	Huangpu
3	F	22	Liwan	36	M	22	Baiyun
4	F	21	Yuexiu	37	F	20	Liwan
5	F	24	Liwan	38	F	24	Liwan
7	F	23	Tianhe	39	M	28	Haizhu
8	F	21	Baiyun	40	F	27	Nansha
10	F	18	Tianhe	41	F	23	Panyu
11	M	21	Yuexiu	43	F	29	Huangpu
12	F	23	Liwan	44	M	22	Huadu
13	F	22	Tianhe	45	F	32	Panyu
14	F	24	Conghua	46	M	30	Yuexiu
15	F	20	Conghua	47	M	23	Haizhu
17	M	22	Tianhe	48	M	27	Yuexiu
18	F	21	Huangpu	49	M	24	Nansha
19	F	23	Panyu	50	M	23	Conghua
20	F	23	Conghua	51	M	23	Baiyun
21	F	21	Tianhe	52	M	26	Yuexiu
22	F	23	Tianhe	53	M	20	Huadu
23	M	19	Tianhe	54	F	32	Tianhe
24	M	20	Liwan	55	M	23	Yuexiu
25	M	20	Yuexiu	56	M	21	Zengcheng
26	F	25	Yuexiu	57	M	32	Haizhu
27	F	25	Tianhe	58	M	26	Haizhu
28	M	25	Tianhe	59	M	29	Liwan
29	M	22	Baiyun	60	M	21	Yuexiu
30	M	25	Haizhu	62	F	23	Liwan
31	F	21	Haizhu	63	M	23	Liwan
32	M	26	Panyu	64	M	24	Liwan
33	F	21	Baiyun	65	M	26	Yuexiu

APPENDIX B: WORD LIST

Appendix B presents the syllables used in the production and perception experiments (Table VI).

TABLE VI. Word list.

	T3(/33/)		T4(/21/)		T6(/22/)	
Syllable	Chinese	English gloss	Chinese	English gloss	Chinese	English gloss
si	试	try	时	time	事	matter
fu	富	rich	扶	support	父	father
sy	恕	forgive	殊	different	竖	vertical
se	泻	leak	蛇	snake	射	shoot
wɐi	畏	fear	遗	lost	胃	stomach
sɐu	秀	show	仇	revenge	寿	longevity
jɐu	幼	young	柔	soft	右	right
jiu	要	need	遥	distant	耀	shine
soey	岁	age	垂	hang	睡	sleep
jim	厌	dislike	炎	flame	验	verify
fan	泛	broad	凡	ordinary	饭	meal
fɐ5n	训	train	坟	grave	份	part
jɐn	印	seal	人	person	孕	pregnant
soen	迅	swift	纯	pure	顺	obey
hoen	汉	Han	寒	cold	汗	sweat
jin	燕	swallow	燃	burn	现	current
jun	怨	complain	原	origin	愿	wish
jeŋ	应	respond	迎	welcome	认	recognize
huŋ	控	control	虹	rainbow	讧	dispute

APPENDIX C: PAIRWISE COMPARISONS OF CLUSTERS FOR EACH TONEPAIR (LOG ODDS RATIO)

Appendix C reports pairwise comparisons between clusters for different tone pairs (Table VII).

TABLE VII. Pairwise comparisons of clusters for each Tone Pair (log odds ratio).

TonePair	Contrast	Estimate	SE	df	z.ratio	p.value
T3–T4	Cluster0–Cluster1	−0.211	0.436	Inf	−0.483	0.9628
	Cluster0–Cluster2	−0.878	0.470	Inf	−1.870	0.2410
	Cluster0–Cluster3	−0.102	0.550	Inf	−0.186	0.9977
	Cluster1–Cluster2	−0.668	0.502	Inf	−1.331	0.5433
	Cluster1–Cluster3	0.109	0.577	Inf	0.188	0.9976
	Cluster2–Cluster3	0.776	0.599	Inf	1.295	0.5658
T3–T6	Cluster0–Cluster1	0.513	0.257	Inf	1.998	0.1887
	Cluster0–Cluster2	0.199	0.252	Inf	0.789	0.8593
	Cluster0–Cluster3	0.397	0.337	Inf	1.179	0.6399
	Cluster1–Cluster2	−0.315	0.253	Inf	−1.245	0.5979
T3–T6	Cluster1– Cluster3	−0.116	0.337	Inf	−0.344	0.9860
	Cluster2–Cluster3	0.198	0.332	Inf	0.597	0.9330
T4-T6	Cluster0–Cluster1	−0.057	0.345	Inf	−0.165	0.9984
	Cluster0–Cluster2	−1.597	0.459	Inf	−3.478	0.0028
	Cluster0–Cluster3	−0.395	0.483	Inf	−0.816	0.8468
	Cluster1–Cluster2	−1.540	0.473	Inf	−3.256	0.0062
	Cluster1–Cluster3	−0.338	0.497	Inf	−0.679	0.9050
	Cluster2–Cluster3	1.203	0.578	Inf	2.081	0.1594

Arnold, L. (**2015**). "Multiple mergers: Production and perception of three pre-/l/ mergers in Youngstown, Ohio," Univ. Pennsylvania Work. Papers Linguistics **21**(2), 2.

Baranowski, M. (**2013**). "On the role of social factors in the loss of phonemic distinctions1," English Lang. Ling. **17**(2), 271–295.

Bates, D., Mächler, M., Bolker, B., and Walker, S. (**2015**). "Fitting linear mixed-effects models using lme4," J. Stat. Softw. **67**(1), 1–48.

Bauer, R. S., Cheung, K.-H., and Cheung, P.-M. (**2003**). "Variation and merger of the rising tones in Hong Kong Cantonese," Lang. Var. Change **15**(2), 211–225.

Blythe, R. A., and Croft, W. (**2012**). "S-curves and the mechanisms of propagation in language change," Language **88**(2), 269–304.

Boersma, P. (**2001**). "PRAAT, a system for doing phonetics by computer," Glot. Int. **5**(9), 341–345.

Bowie, D. (**2000**). "Perception and production in processes of merger," in *Proceedings of the Deseret Language and Linguistic Society Symposium*, Vol. 26, p. 18.

Chao, Y.-R. (**1930**). "A system of tone letters," Le Maître phonétique **45**, 24–27.

Chen, L. (**2011**). "The imposition of Cantonese on Mandarin in the city of Guangzhou," in *Proceedings of the 23rd North American Conference on Chinese Linguistics (NACCL-23)*, Vol. 2.

Cheng, L. S., Babel, M., and Yao, Y. (**2022**). "Production and perception across three Hong Kong Cantonese consonant mergers: Community-and individual-level perspectives," Lab. Phonol. **13**(1), 14.

Freeman, V. (**2014**). "Bag, beg, bagel: Prevelar raising and merger in Pacific Northwest English," U. Wash. Work. Papers Ling **32**, 1–23.

Freeman, V. (**2023**). "Production and perception of prevelar merger: Two-dimensional comparisons using Pillai scores and confusion matrices," J. Phon. **97**, 101213.

Fung, R. S., and Lee, C. K. (**2019**). "Tone mergers in Hong Kong Cantonese: An asymmetry of production and perception," J. Acoust. Soc. Am. **146**(5), EL424–EL430.

Fung, R. S., and Wong, E. Y. (**2023**). "Separated and reunified: An apparent time investigation of the voice quality differences between Hong Kong Cantonese and Guangzhou Cantonese," PLoS One **18**(10), e0293058.

Gardner, M. H., and Roeder, R. V. (**2022**). "Phonological mergers have systemic phonetic consequences: Palm, trees, and the Low Back Merger shift," Lang. Var. Change **34**(1), 29–52.

Grosvald, M., and Corina, D. P. (**2012**). "The production and perception of Sub-phonemic vowel contrasts and the role of the listener in sound change," in *The Initiation of Sound Change: Perception, Production, and Social Factors* (John Benjamins Publishing Company, Amsterdam, the Netherlands), pp. 77–100.

Janson, T. (**1983**). "Sound change in perception and production," Language **59**(1), 18–34.

Johnson, D. E. (**2007**). *Stability and Change along a Dialect Boundary: The Low Vowel Mergers of Southeastern New England* (American Dialect Society, Seattle, WA).

Kleber, F., Harrington, J., and Reubold, U. (**2012**). "The relationship between the perception and production of coarticulation during a sound change in progress," Lang. Speech **55**(3), 383–405.

Kpogo, F. (**2023**). "Age and sex effects on sound change: One size does not fit all," J. Acoust. Soc. Am. **153**, A290–A290.

Labov, W. (**1990**). "The intersection of sex and social class in the course of linguistic change," Lang. Var. Change **2**(2), 205–254.

Labov, W. (**1994**). *Principles of Linguistic Change Principles of Linguistic Change, Internal Factors* (Wiley, New York).

Labov, W., Yaeger, M., and Steiner, R. (**1972**). "A quantitative study of sound change in progress," Report to the National Science Foundation (U.S. Regional Survey, Philadelphia, PA).

Law, S.-P., Fung, R. S.-Y., and Kung, C.-Y. (**2013**). "An ERP study of good production vis-à-vis poor perception of tones in Cantonese: Implications for top-down speech processing," PLoS One **8**(1), e54396.

Lawson, E., Scobbie, J. M., and Stuart-Smith, J. (**2013**). "Bunched/r/promotes vowel merger to schwar: An ultrasound tongue imaging study of Scottish sociophonetic variation," J. Phonetics **41**(3-4), 198–210.

Lenth, R. V. (**2025**). "emmeans: Estimated marginal means, aka least-squares means," https://CRAN.R-project.org/package=emmeans (Last viewed October 31, 2026).

Li, S. (**2008**). "A perception experiment on the yinqu and yangqu tones in Guangzhou Cantonese," Fangyan (Dialect) **01**, 34–39.

Liang, Y. (**2017**). "The production and perception mechanism in tonal variation: A case study of Hong Kong Cantonese," Zhongguo Yuwen (Stud. Chin. Language) **06**, 723–732.
Mok, P. P., Zuo, D., and Wong, P. W. (**2013**). "Production and perception of a sound change in progress: Tone merging in Hong Kong Cantonese," Lang. Var. Change **25**(3), 341–370.
Molina García, Á. (**2023**). "The pitfalls of near-mergers: A sociophonetic approach to near-demergers in the Malaga /θ/ vs /s/ split," Open Linguistics **9**(1), 20220249.
Mufwene, S. (**2008**). *Language Evolution: Contact, Competition and Change* (Bloomsbury Academic, London).
Ohala, J. J. (**1993**). "The phonetics of sound change," in *Historical Linguistics: Problems and Perspectives*, edited by C. Jones (Longman Academic, London), pp. 237–278.
Ou, J. (**2012**). "Tone merger in Guangzhou Cantonese," M. Phil. thesis, The Hong Kong Polytechnic University, Hong Kong.
Ou, J., and Law, S.-P. (**2016**). "Individual differences in processing pitch contour and rise time in adults: A behavioral and electrophysiological study of Cantonese tone merging," J. Acoust. Soc. Am. **139**(6), 3226–3237.
Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., Vanderplas, J., Cournaeau, D., Brucher, M., Perrot, M., and Duchesnay, É. (**2011**). "Scikit-learn: Machine learning in Python," J. Mach. Learn. Res. **12**, 2825–2830.
Python Software Foundation (**2020**). "Python Language Reference, version 3.9," https://www.python.org/ (Last viewed 2025-07-05).
Qin, Z., and Jongman, A. (**2016**). "Does second language experience modulate perception of tones in a third language?," Lang. Speech **59**(3), 318–338.
Qin, Z., and Mok, P. P. (**2013**). "Discrimination of Cantonese tones by speakers of tone and non-tone languages," Kansas Work. Papers Linguistics **34**, 1–25.
R Core Team (**2023**). *R: A Language and Environment for Statistical Computing*, R Foundation for Statistical Computing, Vienna, Austria, https://www.R-project.org/.
Ren, X., and Mok, P. (**2025**). "The effects of contextual tonal variation on Cantonese tone merging," Lab. Phonology: J. Assoc. Lab. Phonology **16**(1), 1–44.
Sautman, B., and Xie, X. (**2020**). "Today in Guangzhou, tomorrow in Hong Kong? A comparative study of the language situation in two cities," J. Curr. Chin. Affairs **49**(2), 207–232.
Thomas, B., and Hay, J. (**2005**). "A pleasant malady: The Ellen/Allan merger in New Zealand English," Te Reo **48**, 69, see https://repository.upenn.edu/server/api/core/bitstreams/aca063d5-ee70-49d3-bd5c-5d1e5d4f7d4a/content.
Thomas, E., and Bailey, G. (**1992**). "A case of competing mergers and their resolution," SECOL Rev. **16**, 179–200.
Venables, W. N., and Ripley, B. D. (**2002**). *Modern Applied Statistics with S* (Springer, New York).
Wang, H., Gao, F., and Zhang, J. (**2024**). "Individual differences in attention and working memory modulate the process of tone merging: Evidence from Macau Cantonese," Humanit. Soc. Sci. Commun. **11**(1), 994.
Wickham, H. (**2016**). "ggplot2: Elegant graphics for data analysis," https://ggplot2.tidyverse.org (Last viewed October 31, 2026).
Wieling, M. (**2018**). "Analyzing dynamic phonetic data using generalized additive mixed modeling: A tutorial focusing on articulatory differences between L1 and L2 speakers of English," J. Phonet. **70**, 86–116.
Wood, S. N. (**2017**). *Generalized Additive Models: An Introduction with R*, 2nd ed. (Chapman and Hall/CRC, London).
Xiong, Z. (**2017**). *A Tutorial on Speech Corpus Construction and Analysis* (Xi'an Jiaotong University Press, Xi'an, China).
Xu, Y., Gandour, J. T., and Francis, A. L. (**2006**). "Effects of language experience and stimulus complexity on the categorical perception of pitch direction," J. Acoust. Soc. Am. **120**(2), 1063–1074.
Yeh, C.-H., and Lin, Y.-H. (**2015**). "Tonal change induced by language attrition and phonetic similarity in hai-lu hakka," in *Prosody and Language in Contact: L2 Acquisition, Attrition and Languages in Multilingual Situations* (Springer, New York), pp. 189–220.
Yu, A. C., and Zellou, G. (**2019**). "Individual differences in language processing: Phonology," Ann. Rev. Linguist. **5**(1), 131–150.
Yu, K. M., and Lam, H. W. (**2014**). "The role of creaky voice in Cantonese tonal perception," J. Acoust. Soc. Am. **136**(3), 1320–1333.
Zebe-Sheng, F., Watter, C., and Schmid, S. (**2025**). "The increasing importance of voice onset time in the perception and production of Zurich German plosives: An ongoing sound change," J. Acoust. Soc. Am. **157**(2), 1261–1275.
Zellou, G. (**2017**). "Individual differences in the production of nasal coarticulation and perceptual compensation," J. Phonetics **61**, 13–29.
Zhan, B. (**2002**). *An Outline of Yue Dialects in Guangdong* (Jinan University Press, Guangzhou, China).
Zhang, J., and Zhu, X. (**2017**). "Tonal chain shifts: A synchronic perspective on tonal system evolution in Huilai and Puning dialects," Zhongguo Yuwen (Chin. Language) **638**(5), 522–535.
Zhu, Q. (**2021**). "Multilingualism in urban China: The case of Guangzhou," in *Multilingual China* (Routledge, London), pp. 30–44.
Zhu, X. (**2004**). "Fundamental frequency normalization: How to handle random variation in tones?," Linguistic Sci. **2**, 3–19.